Fourth-Order Perturbations of Pure-Weyl Black Holes

Exact Ricci–Bach Composition, Horizon Pairing, and Endpoint Nonselection

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Abstract

We ask whether familiar one-ended Lorentzian endpoint conditions isolate the Einstein sector of four-dimensional pure Weyl gravity around a black hole. We first classify the Bach-flat locus in a declared static Laurent ansatz, recover the Mannheim–Kazanas component, and derive a residual-basic horizon generator. On its nondegenerate static domain, the associated Lee–Wald Hamiltonian, Wald entropy, and signed surface-gravity temperature obey an exact first law at every simple horizon. We correct an important sheet distinction: on the complete Laurent locus the metric is Einstein precisely when $\gamma = 0$ and $w = 1$, whereas $\gamma = 0$ alone suffices only on the Mannheim–Kazanas component through $w = 1$.

On Schwarzschild, linearizing the Bach tensor gives an exact Ricci-carrier composition. It defines an exact sequence from Bach solutions to curvature solutions, not a canonical Einstein/additional direct sum. In the axial $\ell = 2$ carrier, $\psi_{ab} = \delta R_{ab}$ obeys a second-order Lichnerowicz-type equation. For nonzero real frequency its horizon system has a two-dimensional analytic ingoing ψ-space, so future-horizon regularity does not force $\psi = 0$. The action-derived Lee–Wald current makes the Einstein self-block isotropic for conjugate Regge–Wheeler waves. Controlled finite-series fixtures at three frequencies exhibit nonzero Einstein–additional cross pairing and nonzero additional-sector pairing; these fixture results are not promoted to a symbolic all-frequency theorem.

At infinity the earlier $\ell = 2$, $M_{\mathrm{BH}}\omega = 3/5$ claim that finite radial pairing selects exactly the Einstein sector is superseded. A Phase-3 audit found that the axial metric reconstruction had also omitted the independent $v\varphi$ Ricci equation. On the declared $\ell = 2$, $M_{\mathrm{BH}} = 1$, $\omega \in [1/2, 3/4]$ pilot this equation propagates as $C' = -2C/r$; its zero fibre is an exact six-dimensional formal module. The repaired non-Einstein X_0 direction is radially finite in the fixed representative and under the true rate-zero Einstein shear, but its cross pairing with the true oscillatory Einstein shear diverges. Thus no representative-independent axial selection theorem follows. Independently, the polar restriction-stable module contains a mixed Einstein/additional line that is finite and generically nonradical, with an explicit algebraic exceptional locus. Thus neither leading endpoint diagnostics nor finite formal radial pairing isolates the Einstein image. Separately, on the same axial pilot, the exact three-dimensional L^2 wave-packet trace spaces at $\mathscr{I}^-$ and $\mathscr{I}^+$ carry action-derived endpoint flux Grams of rank three, zero radical, and inertia $(1, 2, 0)$ for $\alpha_{\mathrm{W}} > 0$, with no frequency wall on the pilot interval. An exact differential-module reduction further gives two spin-two Regge–Wheeler factors and one spin-one Regge–Wheeler factor. Their real-potential Wronskians prove that the global map from future-horizon-regular data to the incoming $\mathscr{I}^-$ trace is invertible for every real $\omega > 0$. Hence the complete indefinite incoming endpoint space is globally populated; its spin-two extension is hyperbolic and its orthogonal spin-one quotient line

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is negative. On the real-frequency cell $0.49995 \le \omega \le 0.50005$, certified uniform nonzero spin-one and spin-two reflection amplitudes and boundary dévissage prove that the complete outgoing trace is also globally populated. Holomorphy then gives outgoing population off a locally finite positive-real exceptional set and dense range on every compact positive-frequency band. Its explicit reflection matrix, the exceptional set, complete-pilot pointwise outgoing rank, non-real poles, a complete asymptotically flat scattering operator, full time-domain stability, damped quasinormal spectrum, Hawking state, and quantum unitarity remain open. The exact filtration does, however, exclude exponentially growing axial $\ell = 2$ separated modes. The local Cauchy restriction $\psi|_\Sigma = \nabla_n\psi|_\Sigma = 0$ still selects the Einstein kernel at the linear level.

AI authorship and computational accountability

The named AI systems developed the programme, derivations, symbolic implementations, independent verification architecture, claim audits, and this manuscript. Asger Alstrup Palm commissioned and coordinated the work and is the corresponding human contact. Every certified statement below is bound to a machine-readable certificate and a separately implemented verifier. The cross-flux result uses controlled high-precision series evaluation with exact recurrence data and independent frequencies; it is reported as fixture-level evidence, not interval-certified nonvanishing. All local tensor identities and polynomial classifications are exact. The manuscript remains conditional on final human review.

Contents

1 The physical question

Fourth-order gravitational equations contain more local solutions than the Einstein equations. A common proposal is to retain the Einstein branch by a boundary condition. Around a one-ended Lorentzian black hole, a first test is:

> Do future-horizon analyticity and the same leading outer admissibility tests used for Einstein waves force the additional Ricci carrier to vanish?

The answer in the certified Schwarzschild axial $\ell = 2$ mode laboratory is no: the curvature carrier has regular ingoing horizon data, the leading real-frequency symbol does not separate it from the Einstein carrier, and controlled fixtures have nonzero action-derived horizon pairing. This is an endpoint-admissibility result, not a classification of all local differential boundary conditions. In particular, setting the Ricci carrier and its normal derivative to zero on a Cauchy surface is a local linear Einstein-sector restriction. Whether that restriction is natural, interaction-stable, or compatible with a physical asymptotic phase space is open.

1.1 Claims and dependency tags

The static family and local operator identities carry LOCAL-ALGEBRAIC. Harmonic branch, horizon, current, and asymptotic statements also carry REDUCED-MODE. The real-frequency incoming population, real-cell and generic outgoing population, compact-band dense-range theorem, boundary dévissage, and separated-mode stability statements additionally carry LORENTZIAN-CAUSAL, each with its declared reduced-mode scope. This does not supply a complete exterior Green-hyperbolicity theorem, metric-falloff phase space, or classified set of local differential boundary operators.

2 Relation to existing branch-selection programmes

The static Bach-flat family itself is classical [2, 3]. Conformal-gravity thermodynamics with additional spin-two hair has also been studied in asymptotically AdS and Einstein–Weyl settings [14]. Our static contribution is therefore not a new family: it is the exact residual-basic normalization and simultaneous signed first-law identity within a declared local quotient. A physical asymptotic normalization has not yet been selected. In particular, Lü, Pang, Pope, and Vázquez-Poritz organize AdS thermodynamics with an independent massive-spin-two-hair conjugate pair. Our one-term identity instead follows after quotient-basic normalization on a restricted static parameter family. The different boundary ensemble, generator normalization, and independent variables mean that the two first laws are complementary, not contradictory.

Maldacena's Einstein-from-conformal construction selects an Einstein sector by Euclidean AdS or late-time de Sitter boundary data [8]. Critical- and logarithmic-gravity analyses similarly separate massless, massive, and generalized modes by asymptotic conditions [9, 13, 10, 11, 12]. The null Einstein self-pairing and nonzero cross pairing found below should be read in that established degenerate-operator context, rather than as the first appearance of null massless modes paired with generalized solutions. Our different question is whether one-ended Lorentzian Schwarzschild endpoint tests force the Ricci carrier to vanish.

Using δR_{ab} to expose higher-derivative spin-two modes is also a known strategy in fourth-order gravity [15]. Previous Weyl-black-hole quasinormal analyses often deliberately obtained a second-order master equation by conformal transport rather than retaining the full fourth-order Bach solution space [16]. Effective-Einstein perturbations of conformally related nonsingular metrics [17] and scalar or electromagnetic test-field spectra [18] are different problems: test-field stability is not stability of the dynamical fourth-order Bach sector. Contemporary quadratic-gravity ringdown work finds both ordinary and additional spin-two sectors [19], while time-domain quadratic-gravity studies track massive-sector tails [20]. The prospective advance here is narrower: the exact pure-Weyl Ricci–Bach sequence, its action-derived Schwarzschild horizon pairing, and failure of the tested endpoint conditions to imply $\delta R_{ab} = 0$. The first-order-system analysis follows the general modified-gravity asymptotic methodology of [21]; in particular, that methodology makes the unresolved repeated-root Jordan audit an essential next step rather than a technical detail. Precise decay of the metric coefficients must ultimately be compared with the Schwarzschild Einstein benchmark [22], not inferred from bounded curvature carrier variables alone.

3 Pure Weyl gravity and its covariant current

We use signature $(-+++)$ and action

$$S_W[g] = \alpha \int_M \sqrt{-g}\, C_{abcd}C^{abcd}\, \mathrm{d}^4x. \tag{1}$$

The field equation is the Bach equation $B_{ab} = 0$, with convention

$$B_{ab} = \nabla^c\nabla^d C_{acbd} + \frac{1}{2}R^{cd}C_{acbd}. \tag{2}$$

The covariant presymplectic current is obtained from the literal variation of the curvature-momentum potential of this action. Its normalization is fixed by the off-shell Green identity

$$\partial_t F^t(h_A, h_B) + \partial_r F^r(h_A, h_B) = 4\alpha \int_{S^2} \sqrt{q}\, \big(h_B^{ab}\delta B_{ab}[h_A] - h_A^{ab}\delta B_{ab}[h_B]\big). \tag{3}$$

Thus the current is conserved for pairs of linearized Bach solutions and is not a freely rescaled master-equation Wronskian.

4 Static spherical Bach-flat family

In Schwarzschild gauge,

$$\mathrm{d}s^2 = -B(r)\mathrm{d}t^2 + B(r)^{-1}\mathrm{d}r^2 + r^2\mathrm{d}\Omega_2^2, \tag{4}$$

the declared Laurent ansatz is

$$B(r) = w - \frac{u}{r} + \gamma r - kr^2 + \frac{c_2}{r^2} + c_3 r^3. \tag{5}$$

Theorem 4.1 (Complete Bach-flat locus in the Laurent ansatz)**.** *The metric is Bach-flat if and only if*

$$c_2 = c_3 = 0, \qquad w^2 + 3u\gamma = 1. \tag{6}$$

The Mannheim–Kazanas parametrization is recovered by

$$w = 1 - 3\beta\gamma, \qquad u = \beta(2 - 3\beta\gamma).$$

On the complete Laurent locus the metric is Einstein if and only if

$$\gamma = 0, \qquad w = 1. \tag{7}$$

On the Mannheim–Kazanas component through $w = 1$, this reduces to the single condition $\gamma = 0$.

Completeness here refers only to the Laurent ansatz. The certificate does not claim a new proof of the global static spherical classification. The sheet qualification in (7) follows already from the scalar curvature

$$R = 12k - \frac{6\gamma}{r} + \frac{2(1-w)}{r^2}. \tag{8}$$

In particular, the allowed point $(\gamma, w) = (0, -1)$ is not Einstein. The original static certificate's phrase "Einstein iff $\gamma = 0$" is therefore used only on its explicitly parametrized Mannheim–Kazanas sheet; the manuscript does not propagate it to the complementary Laurent sheet.

The form-preserving local Diff $\times$Weyl transformations act as dilation and translation on the cubic

$$Q(x) = -ux^3 + wx^2 + \gamma x - k, \qquad x = r^{-1}.$$

Their orbit rank is two. A single continuous invariant may be taken as

$$J = u^2 \operatorname{disc}(Q),$$

while $\operatorname{disc}(Q) = 0$ is the degenerate-horizon locus. The conformal translation may fail globally where its Weyl factor vanishes; it is used only as a local orbit statement.

An exact non-Einstein three-horizon fixture is

$$(\beta, \gamma, k) = \left(\frac{3}{2}, \frac{12}{19}, \frac{1}{19}\right), \qquad rB(r) = -\frac{1}{19}(r-1)(r-3)(r-8).$$

All curvature invariants are finite at the simple horizons. In ingoing Eddington–Finkelstein coordinates,

$$\mathrm{d}s^2 = -B(r)\mathrm{d}v^2 + 2\mathrm{d}v\mathrm{d}r + r^2\mathrm{d}\Omega_2^2,$$

the metric and Bach equation are regular there.

5 Residual-basic energy and the static first law

For the chart generator ∂_t, the bare static Lee–Wald one-form is not integrable. The obstruction disappears when the generator is normalized consistently with the residual quotient.

Theorem 5.1 (Normalized static generator)**.** *On a connected component with $u \neq 0$, choose the oriented generator*

$$\chi = u\,\partial_t, \qquad u = \beta(2 - 3\beta\gamma). \tag{9}$$

Residual basicness and dilation weight determine the general normalization as $uf(J)\partial_t$. We select $f(J) = 1$. Its Hamiltonian is

$$H = -16\pi\alpha\beta^2 D_2, \qquad D_2 = 9\beta^2\gamma^2 k - \beta\gamma^3 - 12\beta\gamma k + \gamma^2 + 4k. \tag{10}$$

At every simple horizon r_h, the Wald entropy

$$S_h = 64\pi^2\alpha\beta\frac{2 - 3\beta\gamma + \gamma r_h}{r_h} \tag{11}$$

and normalized temperature

$$T_h = \frac{uB'(r_h)}{4\pi} \tag{12}$$

satisfy

$$\mathrm{d}H = T_h\,\mathrm{d}S_h \tag{13}$$

identically modulo $B(r_h) = 0$.

Here T_h is the *signed* surface-gravity temperature appropriate to the oriented first-law identity; the positive Hawking temperature would use $|uB'(r_h)|/(4\pi)$. On a component where $u < 0$, future-directedness uses $-\chi$, flipping both H and T_h and preserving the first law. The locus $u = 0$ is excluded because this normalization degenerates. For another allowed $f(J)$, one has $T_f = f(J)T_h$ and a reparametrized Hamiltonian satisfying $\mathrm{d}H_f = f(J)\mathrm{d}H$; the first law remains true. Thus residual basicness fixes a class, not a unique physical normalization in the absence of an asymptotic clock.

This is a theorem on the static parameter slice, not yet a general physical-process first law. Its linear dynamical extension nevertheless passes a strong gauge audit: arbitrary time-dependent spherical Weyl rescalings and diffeomorphisms have exactly zero charge and flux, the entropy is conformally invariant on the symbolic family, and the normalized generator extension is unique at linear order under the declared boundary scalar rule.

6 Ricci–Bach composition and the axial exact sequence

The parity-specific formulas are restrictions of one Ricci-flat identity.

Proposition 6.1 (Linearized Bach tensor on a Ricci-flat background)**.** *For a perturbation of a four-dimensional Ricci-flat metric,*

$$\delta B_{ab} = \frac{1}{2}\Box(\delta R_{ab}) + C_{acbd}(\delta R)^{cd} - \frac{1}{6}\nabla_a\nabla_b\delta R - \frac{1}{12}g_{ab}\Box\delta R. \tag{14}$$

Proof. Rewrite the four-dimensional Bach tensor in Ricci variables using the contracted Bianchi identity, then retain the terms linear in Ricci curvature about $R_{ab} = 0$. Background derivatives and the curvature commutator give the displayed Weyl action. The repository verifies (14) independently, component by component, on the complete axial and polar $\ell = 2$ Regge–Wheeler-gauge carriers. That computer check is carrier-complete but is not presented as a machine proof for arbitrary harmonic content. □

We now specialize the background to Schwarzschild and use axial Regge–Wheeler gauge,

$$h_{t\phi} = h_0(t,r)S_{\ell m}, \qquad h_{r\phi} = h_1(t,r)S_{\ell m}.$$

The $\ell = 2$ linearized Bach tensor is fourth order, trace-free, and obeys the exact linearized Noether identity. Eliminating the constraint on the Einstein subbranch reproduces the Regge–Wheeler equation for $\Psi = Bh_1/r$,

$$B(B\Psi')' + (\omega^2 - V_{\mathrm{RW}})\Psi = 0, \qquad V_{\mathrm{RW}} = B\left(\frac{6}{r^2} - \frac{6m}{r^3}\right). \tag{15}$$

Corollary 6.2 (Axial composition)**.** *For the certified axial carrier, parity gives $\delta R = 0$, so*

$$\boxed{\delta B_{ab} = \frac{1}{2}\Box(\delta R_{ab}) + C_{acbd}(\delta R)^{cd}.} \tag{16}$$

Consequently the Einstein kernel $\delta R_{ab} = 0$ injects into the Bach kernel, while its curvature image obeys

$$\frac{1}{2}\Box\psi_{ab} + C_{acbd}\psi^{cd} = 0, \qquad \psi_{ab} := \delta R_{ab}. \tag{17}$$

Let $\mathcal{C}_{\mathrm{ax}}$ be the fixed axial carrier and define

$$\mathcal{H}_B = \{h \in \mathcal{C}_{\mathrm{ax}} : \delta B[h] = 0\}, \qquad \mathcal{H}_E = \{h \in \mathcal{H}_B : \delta R[h] = 0\}, \tag{18}$$

$$\mathcal{H}_R = \ker\left(\tfrac{1}{2}\Box + C\circ\right) \cap \operatorname{im}(\delta R|_{\mathcal{C}_{\mathrm{ax}}}). \tag{19}$$

Then the invariant statement supplied by the composition is the exact sequence

$$0 \longrightarrow \mathcal{H}_E \longrightarrow \mathcal{H}_B \xrightarrow{\delta R} \mathcal{H}_R \longrightarrow 0. \tag{20}$$

It does *not* supply a canonical direct sum. A metric lift h_X of $\psi \in \mathcal{H}_R$, when it exists in the chosen domain, may be shifted by any $h_E \in \mathcal{H}_E$. Global lift existence, boundary conditions, and a gauge-independent splitting must therefore be proved separately. In particular, "additional branch" below refers either to the curvature image $\mathcal{H}_R$ or to an explicitly constructed lift, never to an assumed canonical complement.

The identity is componentwise and off shell in the perturbation. It does not extend in this two-term form to the certified non-Einstein static fixture. Thus the clean composition is a Ricci-flat-background theorem.

7 The Ricci carrier reaches the future horizon

In ingoing coordinates (v, r), Fourier modes are written $e^{i\omega v}$. The extra carrier's divergence constraint solves its third component polynomially; the apparent Schwarzschild-coordinate pole is a chart artifact.

Theorem 7.1 (Analytic ingoing curvature family)**.** *For axial $\ell = 2$ and $\omega \in \mathbb{R} \setminus \{0\}$, the future horizon $r = 2m$ is a regular singular point of the extra first-order radial system. Its residue spectrum is*

$$\{0, 0, -4im\omega, -2 - 4im\omega\}, \tag{21}$$

and the zero eigenspace has dimension two. Hence there is a two-parameter family of ψ-solutions analytic at the future horizon, with no leading logarithm at the zero exponent.

The restriction excludes $\omega = 0$, where the algebraic multiplicity of the zero residue exponent changes, and makes no claim at resonant complex frequencies. Most importantly, the theorem counts the curvature carrier, not a quotient of metric solutions. The currently certified dimensions are:

Space	Ingoing/analytic dimension	Status
Einstein Regge–Wheeler carrier	1	second-order control
Ricci carrier ψ	2	exact residue calculation
Full Bach metric solution space	not certified	reconstruction/domain open
Metric quotient by the Einstein kernel	not certified	lift and gauge open

Thus future-horizon regularity does not force $\psi = 0$. Explicit metric lifts are constructed in the flux fixtures below, but the exact sequence (20) has not yet been split globally.

8 Action-derived horizon flux

Substituting two on-shell Regge–Wheeler solutions into the current (3) gives

$$F^r_{EE} = -\frac{192\pi\alpha(\omega_1^2 - \omega_2^2)}{5\omega_1\omega_2 r}\Psi_1\Psi_2. \tag{22}$$

For conjugate pairs $\omega_2 = \pm\omega_1$, this vanishes identically.

Theorem 8.1 (Exact Einstein isotropy)**.** *On Schwarzschild in axial $\ell = 2$, the pure-Weyl Lee–Wald pairing of an Einstein Regge–Wheeler mode with its conjugate vanishes exactly:*

$$F^r(E_\omega, \overline{E_\omega}) = 0. \tag{23}$$

Thus the Einstein line is isotropic in the presymplectic structure inherited from the pure C^2 action. This statement does not identify the Einstein symplectic form with the ordinary Einstein–Hilbert one.

Proposition 8.2 (Controlled horizon-pairing fixtures)**.** *For $m = 1$, axial $\ell = 2$, and conjugate-frequency Hermitian pairings, order-16 ingoing series give*

$$F^r(E_\omega, \overline{X_\omega}) \neq 0, \qquad F^r(X_\omega, \overline{X_\omega}) \neq 0 \tag{24}$$

at producer frequencies $\omega = 3/5, 2/7$; an independent verifier repeats the nonvanishing gates at $\omega = 1/2$. At the two stored frequencies the values, divided by $\pi\alpha$ and sampled at two interior radii, are approximately

ω	$F^r(X,\bar{X})/(\pi\alpha)$	$F^r(E,\bar{X})/(\pi\alpha)$
3/5	$-330.58i, -329.98i$	$-218.17 + 43.55i, -215.97 + 51.53i$
2/7	$-71.92i, -71.95i$	$-162.82 + 27.99i, -163.92 + 33.03i$

The exact Regge–Wheeler null identity is the in-run control. These data are controlled numerical fixtures: they prove neither symbolic nonvanishing for all real ω nor an interval-certified error bound.

For a chosen lift basis (E, X), the horizon block has the form

$$\Omega_h = \begin{pmatrix} 0 & a \\ -\bar{a} & b \end{pmatrix}. \tag{25}$$

Under the admissible lift change $X \mapsto X + cE$, the cross entry a is invariant while b changes. The robust fixture-level conclusions are therefore Einstein isotropy, nonzero cross pairing, and full rank of the tested two-dimensional block. The sign of iF^r_{XX} for one chosen lift is basis dependent and is not promoted to a branch signature or Hilbert norm.

The stored report also records a repaired calculation: an earlier flux evaluation dropped the Fourier factor $e^{i\omega t}$, causing all time derivatives to vanish silently. The exact Regge–Wheeler null control exposed the defect; all displayed fixtures use the repaired full-time current.

9 Asymptotic symbol and endpoint nonselection

At real frequency and large radius, the extra carrier has dispersion

$$(\lambda_r^2 - \omega^2)^2 = 0 \tag{26}$$

and growth exponents

$$\sigma \in \{\pm 2im\omega,\ \pm 2im\omega - 1\}. \tag{27}$$

The simple-root reading would identify Coulomb logarithmic phases with amplitudes of order r^0 and r^{-1}. But the characteristic is *double*. The polynomial alone does not determine geometric multiplicity. As in $(D^2 + \omega^2)^2 h = 0$, a Jordan chain may produce $r_* e^{\pm i\omega r_*}$ even though every characteristic exponent is oscillatory. The axial Ricci-carrier recurrences are known to all formal orders for every integer $\ell \geq 2$ and real $\omega \neq 0$. The earlier generic-ℓ metric lifts, however, did not impose the independent $v\varphi$ Ricci row. The complete metric endpoint module is certified here only at $\ell = 2$ on the real pilot interval. In the polar calculation below we use only the maximal restriction-stable module through the certified depth. Two additional terminal-only prefix vectors still require their next compatibility test and are not promoted to formal solutions.

Theorem 9.1 (Endpoint regularity-and-leading-symbol nonselection)**.** *On the declared Schwarzschild axial $\ell = 2$, nonzero-real-frequency mode carrier, future-horizon analyticity does not imply $\delta R_{ab} = 0$. Moreover, the presently certified leading characteristic polynomial and formal exponents do not supply an outer condition that distinguishes the Ricci carrier from the Einstein characteristic. Hence these tested local endpoint diagnostics do not select the Einstein kernel.*

This deliberately does *not* say that every ordinary asymptotically flat radiation condition admits an additional metric mode. That stronger statement requires convergence or another completion of the formal series, falloff in a regular tetrad or asymptotic gauge, a differentiable Lee–Wald phase space, and a horizon-to-infinity connection map. None is supplied by the formal calculations below.

9.1 Complete axial reconstruction on the $\ell = 2$ pilot

The earlier generic-angular metric audit imposed the $x\varphi$ and $r\varphi$ reconstruction equations but omitted the independent $v\varphi$ equation. Put

$$C = \frac{\delta R_{v\varphi}}{S_2} - P. \tag{28}$$

On every solution of the two differential rows and the Ricci-carrier equations, exact differentiation gives

$$C' = -\frac{2}{r}C, \qquad \kappa = r^2 C = \text{constant}. \tag{29}$$

The complete reconstruction is therefore the $\kappa = 0$ fibre. Solving this algebraic constraint for H_0 produces an exact six-state flow on $(P, P', Q, Q', H_1, H_1')$ and the short exact sequence

$$0 \longrightarrow \mathcal{E}^2_{\mathrm{ker}} \longrightarrow \mathcal{E}^6_{\mathrm{Bach,ax}} \longrightarrow \mathcal{E}^4_{\mathrm{Ricci}} \longrightarrow 0. \tag{30}$$

It supplies six independent formal columns at each endpoint: four carrier lifts and two complete Einstein-kernel columns. This statement is exact for $\ell = 2$, $M = 1$, and real $\omega \in [1/2, 3/4]$; it is not a generic-ℓ metric theorem.

The legacy Phase-2 X_0 has

$$C_{X_0} = \frac{i(\omega - 18i)}{2\omega^2} r^{-2}. \tag{31}$$

It becomes a complete non-Einstein lift after an exact transverse addition. The vector formerly labelled E_0 is one half of that transverse vector and is not itself a complete Einstein solution; all claims using it as an Einstein representative are superseded.

For the repaired X_0, the literal current audit has no coefficient at any nonintegrable power $p \geq -1$. Its first self coefficient is

$$F^v(X_0, \overline{X}_0) = \frac{32 i\pi\alpha_{\mathrm{W}}(540 - \omega^2)}{15\omega^3(\omega^2 + 4)} r^{-2} + O(r^{-3}), \tag{32}$$

which is nonzero on the pilot interval. The same finite class is preserved under the true rate-zero Einstein-kernel shear. It is *not* preserved under an arbitrary Einstein shear: the true oscillatory Einstein kernel has cross term

$$F^v(E_{\mathrm{osc}}, \overline{X}_0) = \frac{48\pi\alpha_{\mathrm{W}}\omega^3(4\omega + i)}{5} e^{-2i\omega r} r^{3-4i\omega} + \cdots, \tag{33}$$

with a nonzero coefficient throughout the real pilot interval.

Theorem 9.2 (Scoped axial reconstruction and radial disposition)**.** *For $\ell = 2$, $M = 1$, and real $\omega \in [1/2, 3/4]$, the complete axial formal endpoint module has dimension six. Its repaired X_0 direction is non-Einstein and radially finite in the fixed representative and under true rate-zero Einstein shears, while true oscillatory Einstein shears produce divergent cross terms. Thus the pilot supplies a finite formal non-Einstein direction but no representative-independent axial selection or quotient theorem.*

This is an exact formal asymptotic coefficient audit. It establishes no convergence or endpoint wave-packet flux space *by itself*, and it establishes no horizon-to-infinity matching, global scattering channel, stability, or CPT property. The endpoint flux construction below is a separate completion of one part of that boundary problem. The other axial additional direction and the generic angular tower require separate all-row audits.

9.2 Exact axial null-endpoint wave-packet flux

The complete axial pilot admits exact matching trace spaces

$$\mathcal{X}_{\mathscr{I}^-} = \mathrm{span}\{\mathrm{XI0}, \mathrm{XI1}, \mathrm{EI0}\}, \qquad \mathcal{X}_{\mathscr{I}^+} = \mathrm{span}\{\mathrm{XI2}, \mathrm{XI3}, \mathrm{EI2}\}. \tag{34}$$

Positive-frequency amplitudes lie initially in $C_c^\infty((1/2, 3/4); \mathbb{C}^3)$, with negative frequencies fixed by the real-field involution; the completion is $L^2([1/2, 3/4]; \mathbb{C}^3)$ at each endpoint. Pulling the frozen action-derived current from Schwarzschild t coordinates to the ingoing Eddington–Finkelstein reconstruction, including differentiated reconstruction terms, gives Stokes-oriented Hermitian Grams G_- and G_+. After division by $\pi\alpha_{\mathrm{W}}$,

$$\det G_- = \frac{14155776}{125}\omega^3, \qquad \det G_+ = \frac{3538944}{125}\omega. \tag{35}$$

The past sign includes the reversal between the coordinate-radial direction and the outward Stokes orientation at $\mathscr{I}^-$.

Theorem 9.3 (Scoped axial null-endpoint flux)**.** *For strict pure Weyl gravity with $M = 1$, axial $\ell = 2$, and $\omega \in [1/2, 3/4]$, both exact wave-packet endpoint trace spaces have dimension three. Their action-derived Stokes Grams have rank three, zero radical, no frequency wall, and*

$$\operatorname{inertia}(G_-) = \operatorname{inertia}(G_+) = (1, 2, 0) \qquad (\alpha_{\mathrm{W}} > 0).$$

Reversing the overall action sign reverses the positive and negative counts.

The signature has a simple exact anatomy. With the displayed Stokes orientations and after division by $\pi\alpha_{\mathrm{W}}$, at $\mathscr{I}^-$ set

$$E_- = \mathrm{EI0}, \qquad X_- = \mathrm{XI0}, \qquad Y_- = -\omega^2\mathrm{XI0} - 2i\omega\mathrm{XI1} + \mathrm{EI0}.$$

Then

$$G_-(E_-, E_-) = 0, \quad G_-(E_-, X_-) = \frac{384\omega}{5}, \quad G_-(Y_-, Y_-) = -\frac{384\omega^3}{5},$$

and Y_- is orthogonal to E_- and X_-. Likewise, at $\mathscr{I}^+$ set

$$E_+ = \mathrm{EI2}, \qquad X_+ = \mathrm{XI3}, \qquad Y_+ = i\omega\mathrm{XI2} + 4(4\omega^2 - i\omega - 2)\mathrm{XI3}.$$

Here

$$G_+(E_+, E_+) = 0, \quad G_+(E_+, X_+) = \frac{96\omega}{5}, \quad G_+(Y_+, Y_+) = -\frac{384\omega}{5},$$

with the same orthogonality. The Einstein–additional planes have determinants $-147456\omega^2/25$ and $-9216\omega^2/25$, respectively. Thus each endpoint has the exact Witt decomposition

$$(1, 2, 0) = (1, 1, 0) \oplus (0, 1, 0).$$

This identifies a hyperbolic Einstein–additional plane and a separate negative line. It does not identify the additional column as a spectral derivative or time-translation Jordan partner. Indeed, the complete six-dimensional axial module is not conjugate by an endpoint-regular invertible change of variables to a scalar Regge–Wheeler square: the latter has only four local columns, whereas the certified exact sequence has dimensions $2 + 4 = 6$. Exact cyclic elimination of the four-dimensional Ricci carrier instead gives

$$L_4 = L_x \circ L_{\mathrm{RW}},$$

and exact embedding and quotient maps certify

$$0 \longrightarrow M_{\mathrm{RW}} \longrightarrow M_{A_4} \longrightarrow M_x \longrightarrow 0.$$

The carrier-zero metric kernel is independently equivalent to M_{RW}. After the rational rescaling $Z = y/[r^2(r-2)]$, the quotient equation is

$$D_{r_*}^2 y + 2i\omega D_{r_*} y - \frac{6(r-2)}{r^3} y = 0,$$

the $\ell = 2$ spin-one Regge–Wheeler equation in the ingoing-EF convention. Thus the full module has a certified triangular filtration with diagonal factors $M_{\mathrm{RW}}, M_{\mathrm{RW}}, M_x$. The natural exact gauge retains a nonzero extension term, so a direct $\ker L_{\mathrm{RW}}^2 \oplus \ker L_x$ splitting and a time-translation Jordan interpretation are not claimed.

The incoming quotient map nevertheless resolves the Witt signs by differential factor. In the basis $(\mathrm{XI0}, \mathrm{XI1}, \mathrm{EI0})$, it obeys

$$\pi_x(\mathrm{XI0}) = 2, \qquad \pi_x(\mathrm{XI1}) = -2i\omega, \qquad \pi_x(\mathrm{EI0}) = 0.$$

Set

$$R = \mathrm{XI0} - \frac{i}{\omega}\mathrm{XI1}, \qquad S = -\omega^2\mathrm{XI0} - 2i\omega\mathrm{XI1} + \frac{4}{3}\mathrm{EI0}, \qquad E = \mathrm{EI0}.$$

Then $\ker \pi_x = \mathrm{span}\{E, R\}$, while $\pi_x(S) = -6\omega^2$. In the ordered basis (R, S, E), the exact incoming Gram divided by $\pi\alpha_{\mathrm{W}}$ is

$$\begin{pmatrix} 624/(5\omega) & 0 & 576\omega/5 \\ 0 & -384\omega^3/5 & 0 \\ 576\omega/5 & 0 & 0 \end{pmatrix}.$$

Consequently the metric/carrier spin-two extension has inertia $(1, 1, 0)$, and its orthogonal spin-one quotient lift is strictly negative, of inertia $(0, 1, 0)$. Equivalently, after $R \mapsto R - 13E/(24\omega^2)$, this is the canonical orthogonal sum

$$\begin{pmatrix} 0 & 576\omega/5 \\ 576\omega/5 & 0 \end{pmatrix} \widehat{\oplus} \left(-384\omega^3/5\right).$$

This factor attribution does not require the differential extension to split. With the quotient normalized to one,

$$S_{\mathrm{unit}} = -\frac{S}{6\omega^2}, \qquad \pi_x(S_{\mathrm{unit}}) = 1, \qquad \frac{G_-(S_{\mathrm{unit}}, S_{\mathrm{unit}})}{\pi\alpha_{\mathrm{W}}} = -\frac{32}{15\omega}.$$

In this canonical basis the fiberwise form is congruent to $\mathrm{diag}(1, -1, -1)$. Across the whole positive frequency axis its natural Hilbert majorant is weighted:

$$\int_0^\infty \left[\frac{576\omega}{5}(|e|^2 + |r|^2) + \frac{32}{15\omega}|s|^2\right] d\omega.$$

Thus a finite-flux spin-one amplitude must obey $\int_0^\epsilon |s(\omega)|^2 d\omega/\omega < \infty$ near threshold. On the positive-frequency Fourier core these are the homogeneous fractional Sobolev weights

$$(e, r) \in \dot{H}^{1/2}, \qquad s \in \dot{H}^{-1/2},$$

up to Fourier normalization and the real-field extension. Weighted integrability alone does not define $s(0)$; if a continuous threshold trace is separately available, it must vanish. The unweighted L^2 topology remains equivalent on compact frequency bands away from zero, while the static/memory sector at $\omega = 0$ is separate. Pointwise invertibility of T_- does not by itself give a bounded global operator on this weighted direct integral. In weight-normalized factor frames, write

$$T_- = \begin{pmatrix} a_2 & b & c \\ 0 & a_2 & d \\ 0 & 0 & a_1 \end{pmatrix}, \qquad a_s = A_{\mathrm{in},s}.$$

Since the scalar Wronskians give $|a_s^{-1}| \le 1$ on the positive real axis, boundedness of the incoming-to-horizon population map T_-^{-1} reduces to weighted bounds on only

$$\frac{b}{a_2^2}, \qquad \frac{d}{a_2 a_1}, \qquad \frac{bd - a_2 c}{a_2^2 a_1}.$$

These are not three unrelated functions. Introduce the partially dualized two-channel family

$$C(\tau) = \begin{pmatrix} a_2(\tau) & d(\tau) \\ 0 & a_1 \end{pmatrix}, \qquad a_2'(0) = b, \qquad d'(0) = c,$$

where the spin-one quotient is held τ-constant. In the ordered basis (ϵ spin-2, spin-2, spin-1), its first jet is exactly T_-, and

$$-\frac{b}{a_2^2} = \partial_\tau \frac{1}{a_2(\tau)}\bigg|_0, \qquad -\frac{d}{a_2 a_1} =: \mathcal{M}(0), \qquad \frac{bd - a_2 c}{a_2^2 a_1} = \mathcal{M}'(0),$$

with $\mathcal{M}(\tau) = -d(\tau)/(a_2(\tau)a_1)$. Thus the quantitative gate is a bound for one base spin-two/spin-one transfer matrix and its intrinsic first variation. This algebraic repackaging does not itself provide the analytic τ-family or its uniform threshold, high-frequency, and limiting-absorption bounds.

The normalized signature basis also defines the canonical classical fundamental symmetry $C_{\text{fac}} = \operatorname{diag}(1, -1, -1)$. In the null basis it exchanges EI and RI_0, negates SI_{unit}, and yields the positive majorant $\operatorname{diag}(a, a, b)$. This is an endpoint Krein-space fundamental symmetry, not a Mannheim C operator: BRST descent and compatibility with the full scattering dynamics are not established.

The L^2 completion here uses the auxiliary positive coefficient norm, not the indefinite flux as a norm. Exact Sturm counts applied to $\|G_\pm(\omega)\|_F^2$ and $\|G_\pm(\omega)^{-1}\|_F^2$ give the uniform normalized estimate

$$\|a\|_{L^2} \le \|G_\pm a\|_{L^2} \le 645\,\|a\|_{L^2}. \tag{36}$$

Thus the endpoint forms are bounded and uniformly nondegenerate, rather than merely pointwise invertible. The continuous fundamental symmetry $J_\pm = \operatorname{sign}(G_\pm)$ and positive majorant $|G_\pm|$ make each endpoint a Krein space with topology equivalent to the auxiliary L^2 topology. This majorant is not a CPT metric, positive energy, or particle norm. The exact matrices and separate endpoint bounds are printed in Appendix A.

The past-boundary sign is already absorbed into the incoming Gram. Hence a future global relation $a_{\text{out}} = T a_{\text{in}}$ must obey

$$T^\dagger J_{\text{out}} T = J_{\text{in}}; \tag{37}$$

without horizon channels this would reduce to $T^\dagger G_+ T = G_-$. No complete outgoing-plus-horizon scattering map is constructed here. The theorem below does, however, prove that the full incoming $\mathscr{I}^-$ form is globally populated by future-horizon-regular solutions.

For the presymplectic ambiguity $\theta \mapsto \theta + \delta Y + dZ$, the δY term vanishes by $\delta^2 = 0$, and angular exact terms integrate to zero. A stationary, globally defined local finite-tangential-jet dZ with finite trace-only pullback contributes only cut terms; these vanish on the C_c^∞ frequency core because the inverse Fourier traces are Schwartz, and hence on its continuous L^2 extension. This scoped statement does not cover radial or subleading, nonlocal or soft, explicit-time, or nondecaying improvements. Their additive $\Delta G_\pm$ and inertia strata remain open.

Proposition 9.4 (Index pullback and local wave-packet consequence)**.** *Fix either certified axial endpoint and a frequency at which its Hermitian flux form $G(\omega)$ has inertia $(1, 2, 0)$. Let*

$$T_\omega : U_\omega \longrightarrow \mathcal{X}_\omega$$

be any globally matched, gauge-reduced trace map, and set $K_\omega = T_\omega^\dagger G(\omega) T_\omega$. *After quotienting* $\ker T_\omega$, *write* $r = \operatorname{rank} T_\omega$. *Then*

$$n_-(K_\omega) \geq r - 1.$$

In particular, a populated endpoint image of rank at least two necessarily contains a negative-flux direction, while full rank forces inertia $(1, 2, 0)$.

If T_ω *and the horizon-regular frame vary continuously near an interior frequency* ω_0, *then* $\operatorname{rank} T_{\omega_0} \geq 2$ *implies the existence of a nonzero smooth compact-frequency horizon amplitude whose populated endpoint flux is strictly negative. In fact the negative index of the populated wave-packet space is then infinite.*

Proof. Let $W = \operatorname{im} T_\omega$. If the restriction of G to W has inertia (a, b, c), it has a nonnegative subspace of dimension $a + c$. Projection of such a subspace onto the one-dimensional positive spectral subspace of G is injective: a vector with zero positive projection has nonpositive norm in the negative spectral subspace, and nonnegativity forces it to vanish. Hence $a + c \leq 1$, and $b = r - a - c \geq r - 1$.

For the second statement choose $u \in U_{\omega_0}$ with $u^\dagger K_{\omega_0} u < 0$. In a continuous local trivialization the same coefficient vector has negative flux on some open interval $J \ni \omega_0$. Multiplying it by any nonzero $\chi \in C_c^\infty(J)$ and integrating the diagonal frequency-space pairing gives a strictly negative finite wave-packet flux. Choose countably many nonzero cutoffs with pairwise disjoint supports inside J. Their cross-pairings vanish, while every self-pairing is negative. They therefore span an infinite-dimensional negative subspace. □

The first global gate can in fact be closed analytically for the incoming endpoint. Let

$$T_-(\omega) : \operatorname{span}\{\mathrm{XH0a}, \mathrm{XH0b}, \mathrm{EH0}\} \longrightarrow \operatorname{span}\{\mathrm{XI0}, \mathrm{XI1}, \mathrm{EI0}\}$$

map future-horizon-regular coefficients to incoming $\mathscr{I}^-$ coefficients.

Theorem 9.5 (Global population of the axial incoming trace)**.** *For every real* $\omega > 0$,

$$T_-(\omega) \in GL(3, \mathbb{C}).$$

In factor-adapted endpoint frames its determinant is

$$\det T_-(\omega) = -\frac{(2\omega - i)(4\omega - i)^2}{4(\omega - i)} A_{\mathrm{in},2}(\omega)^2 A_{\mathrm{in},1}(\omega),$$

where $A_{\mathrm{in},2}$ *and* $A_{\mathrm{in},1}$ *are the incoming Jost coefficients for the spin-two and spin-one real short-range potentials*

$$V_2 = \frac{6(r-2)(r-1)}{r^4}, \qquad V_1 = \frac{6(r-2)}{r^3}.$$

Proof. The exact module filtration and endpoint quotient maps give one horizon-ingoing and one incoming-infinity Jost line for each of the three diagonal factors. The changes from the repository endpoint frames to these factor-adapted frames are triangular with determinant one. The resulting connection is triangular and its diagonal entries are the two spin-two and one spin-one Jost connection coefficients, multiplied by the displayed nonzero endpoint normalizations. For a unit horizon-ingoing solution of either real potential, Wronskian conservation gives

$$|A_{\mathrm{in},s}|^2 - |A_{\mathrm{out},s}|^2 = 1, \qquad s = 1, 2.$$

Hence $A_{\mathrm{in},s} \neq 0$. The rational prefactor is also nonzero, since its squared modulus is

$$\frac{(4\omega^2+1)(16\omega^2+1)^2}{16(\omega^2+1)}.$$

The cyclic and triangular-map determinants are nonzero for $r > 2$ and real $\omega > 0$; the horizon recurrence pivots do not vanish, their integer-spaced resonance is compatible, and the two infinity rates remain distinct. Thus no endpoint-map or Frobenius collision interrupts the argument on the positive real axis. The zero-frequency sector is separate. □

Moreover, on the original pilot interval the determinant has the uniform exact margin

$$|\det T_-(\omega)| \geq \sqrt{\frac{5}{2}}.$$

Indeed, after $x = \omega^2$, the derivative of the squared rational prefactor is

$$\frac{(16x+1)(128x^2+208x+35)}{16(x+1)^2} > 0,$$

and $|A_{\mathrm{in},s}| \geq 1$. This is a volume-determinant bound, not by itself a smallest-singular-value estimate. On the full positive real axis the same monotonicity gives

$$|\det T_-(\omega)| > \frac{1}{4}, \qquad \omega > 0,$$

with $1/4$ approached only as $\omega \to 0^+$.

Congruence by the invertible T_- preserves inertia. The horizon-regular pullback of the incoming endpoint form therefore has inertia $(1, 2, 0)$ at every real positive frequency, and its wave-packet completion on every compact positive-frequency interval has infinite positive and negative index. In particular, the conditional conclusion of Proposition 9.4 is activated for $\mathscr{I}^-$: the incoming indefinite directions are genuine globally populated radial channels and cannot be removed by finitely many global charge conditions, a finite-dimensional residual quotient, or deletion of finitely many exceptional frequencies.

This is a one-sided global connection and flux theorem, not yet a complete all-frequency scattering theorem. The real cell certified below determines the rank of the outgoing $\mathscr{I}^+$ reflection block T_+, and analyticity determines it away from a locally finite positive-real exceptional set. The locations and possible existence of those isolated reflection zeros remain open, as do the entries of T_+, the reflection amplitudes, non-real complex-frequency poles, total canonical energy, a particle norm, CPT positivity, time-domain stability, or invariance under unrestricted radial or corner improvements of the presymplectic potential. It does exclude a future-horizon-regular solution with zero incoming $\mathscr{I}^-$ trace, and hence excludes purely outgoing real-frequency axial modes and real-axis incoming spectral singularities for every $\omega > 0$.

There is also a scoped complex-frequency result. The repository phase is $e^{+i\omega t}$, so exponential growth corresponds to $\operatorname{Im}\omega < 0$. A zero of either reduced incoming Jost coefficient in that half-plane would produce an $L^2(dr_*)$ mode of

$$-D_{r_*}^2 + V_s, \qquad s = 1, 2,$$

with eigenvalue ω^2. Both displayed potentials are nonnegative. Self-adjointness excludes a nonreal eigenvalue when $\operatorname{Re}\omega \neq 0$, and nonnegativity excludes $\omega^2 < 0$ on the imaginary axis. The exact inclusion, quotient, and master maps preserve the future-horizon-regular and pure-outgoing germ classes throughout that half-plane. Successive quotient elimination therefore proves that the complete non-split six-state axial $\ell = 2$ system has no growing separated mode. Equivalently, its intrinsic factor Evans product

$$\mathcal{E}_{\mathrm{reg}}(\omega) = A_{\mathrm{in},2}(\omega)^2 A_{\mathrm{in},1}(\omega)$$

is zero-free for $\operatorname{Im}\omega < 0$. This is a separated-mode spectral no-growth theorem, not time-domain boundedness, decay, completeness, or a full PDE stability theorem. The frame events at $\omega = i/4, i/2, i$ lie in the temporally damped half-plane in this convention and remain unclassified by the energy argument.

The non-split spin-two extension can also be extracted exactly in local Regge–Wheeler companion frames. Its rational 2×2 source matrix $\mathcal{E}_{\mathrm{RW}}$ has rank one, but it is not a pointwise scalar multiple of $\partial_\omega A_{\mathrm{RW}}$. More strongly, a putative global rational-gauge identity

$$\mathcal{E}_{\mathrm{RW}} = q(\omega)\partial_\omega A_{\mathrm{RW}} + B' + BA_{\mathrm{RW}} - A_{\mathrm{RW}}B$$

has trace residue $4iq$ at $r = 2$, whereas a rational derivative has zero residue. It therefore forces $q = 0$ in that gauge class. This does not exclude endpoint-analytic logarithmic spectral-phase gauges.

At a simple damped spin-two zero, the invariant scattering datum remains $[b] \in \mathcal{O}/(A_{\mathrm{in},2})$, where b is the off-diagonal entry between the repeated spin-two factors in the triangular connection. The congruence $b \equiv qA'_{\mathrm{in},2} \pmod{A_{\mathrm{in},2}}$ with unspecified q is not predictive: $[A'_{\mathrm{in},2}]$ is a unit, so such a class $[q]$ always exists. Selecting the local Smith type requires a certified QNM and adjoint germ together with a convergent Fredholm pairing; no nonzero extension resonance is promoted here.

Proposition 9.6 (Local QNM Smith dichotomy)**.** *Let ω_n be a simple zero of the spin-two incoming Jost coefficient $a = A_{\mathrm{in},2}$, and suppose the spin-one coefficient $f = A_{\mathrm{in},1}$ is nonzero there. After analytic elimination of the spin-one factor, the local connection is equivalent to*

$$\begin{pmatrix} a & b \\ 0 & a \end{pmatrix}.$$

If $[b] \neq 0$ in $\mathcal{O}_{\omega_n}/(a)$, its Smith valuations are $(0, 2)$: the resonance has one root vector, a length-two chain, and a second-order pole of the inverse connection matrix. A second-order pole of the full differential resolvent is a conditional corollary once the physical QNM boundary problem is realized as the corresponding analytic Fredholm pencil. If $[b] = 0$, the valuations are $(1, 1)$: there are two independent root vectors and only simple poles of the inverse connection matrix.

More explicitly, write $z = \omega - \omega_n$, $a = \alpha_n z + O(z^2)$, and $b = \beta_n + O(z)$. When $\beta_n \neq 0$, the double-pole coefficient is

$$P_{-2} = -\frac{\beta_n}{\alpha_n^2} E_{12},$$

and the generalized root coefficient is controlled by $\kappa_n = \beta_n/\alpha_n$.

At the resonant frequency the full connection has the exact minor

$$\Delta_n = \det T_-[\{1, 3\}, \{2, 3\}] = b(\omega_n) A_{\mathrm{in},1}(\omega_n).$$

Since the spin-one factor is a unit,

$$\beta_n \neq 0 \iff \Delta_n \neq 0 \iff \operatorname{rank} T_-(\omega_n) = 2;$$

the zero-class branch has rank one. This supplies a direct validated-minor test of the binary Smith case without first constructing an adjoint germ. The chosen minor is tied to the normalized triangular factor frame. Invariantly, the second Fitting ideal generated by all 2×2 minors is the unit ideal in the nonzero class and (a) in the zero class; the full Smith valuations are therefore $(0,0,2)$ and $(0,1,1)$, respectively.

Under compatible endpoint normalizations the selector is the invariant Fredholm overlap

$$\alpha_n = \langle \widetilde{u}_n,\, \partial_\omega L_{\mathrm{RW}}(\omega_n) u_n \rangle\,, \qquad \beta_n = \langle \widetilde{u}_n,\, \mathcal{E}_{\mathrm{RW}}(\omega_n) u_n \rangle\,, \qquad [b] \neq 0 \iff \beta_n \neq 0.$$

For an endpoint-preserving analytic triangular change, $\mathcal{E}_{\mathrm{RW}} \mapsto \mathcal{E}_{\mathrm{RW}} + L_{\mathrm{RW}} Q - Q L_{\mathrm{RW}}$, its change in the overlap is

$$\langle \widetilde{u}_n, (L_{\mathrm{RW}} Q - Q L_{\mathrm{RW}}) u_n \rangle = \left\langle L_{\mathrm{RW}}^{\dagger} \widetilde{u}_n, Q u_n \right\rangle - \langle \widetilde{u}_n, Q L_{\mathrm{RW}} u_n \rangle = 0.$$

Thus β_n represents the intrinsic extension class. The present certificate proves this dichotomy and invariance, but does not evaluate β_n at any physical QNM.

More precisely, on any analytic Fredholm realization for which the repeated spin-two operator has block form

$$\mathbb{L}(\omega) = \begin{pmatrix} L(\omega) & \mathcal{E}(\omega) \\ 0 & L(\omega) \end{pmatrix},$$

the exact inverse identity gives

$$\mathbb{L}(\omega)^{-1} = -\frac{\beta_n}{\alpha_n^2} \frac{1}{(\omega - \omega_n)^2} \begin{pmatrix} 0 & u_n \otimes \widetilde{u}_n \\ 0 & 0 \end{pmatrix} + O((\omega - \omega_n)^{-1}).$$

Hence $\beta_n \neq 0$ implies a genuine second-order pole of that full differential resolvent. Generic sources with nonzero adjoint-QNM projection then excite a $te^{i\omega_n t}$ term. The analytic Fredholm realization of the physical QNM boundary problem is not constructed here.

Proposition 9.7 (No Schwarzschild-parameter trivialization)**.** *Reissue the axial spin-two Regge–Wheeler companion $A_{\mathrm{RW}}(r, \omega; M)$ with dimensionful areal radius r, frequency ω, and Schwarzschild mass M. If its rational extension source admits*

$$\mathcal{E}_{\mathrm{RW}} = q_\omega \partial_\omega A_{\mathrm{RW}} + q_M \partial_M A_{\mathrm{RW}} + D_A B, \qquad D_A B = B' + B A_{\mathrm{RW}} - A_{\mathrm{RW}} B,$$

then the horizon and infinity trace residues require

$$q_\omega + \omega q_M = 0.$$

At $M = 1$, however,

$$\partial_M A_{\mathrm{RW}} - \omega \partial_\omega A_{\mathrm{RW}} = D_A(-r A_{\mathrm{RW}} - \mathrm{diag}(0,1))\,.$$

Hence every residue-compatible mass/frequency parameter contribution is rationally exact. Thus the mixed parameter ansatz reduces to the pure rational-coboundary question addressed next; it supplies no independent Schwarzschild-family trivialization.

The proposition by itself does not decide whether $\mathcal{E}_{\mathrm{RW}}$ is a pure rational D_A-coboundary, and it does not evaluate β_n. It excludes the direct interpretation of a future generalized resonance as $\partial_M u_n$, $\partial_\omega u_n$, or their residue-compatible combination. Cosmological-constant variation requires a separately certified Schwarzschild–(A)dS boundary problem.

Proposition 9.8 (Projective cocycle of the spin-two self-extension)**.** *Put the scalar Regge–Wheeler factor in Liouville form*

$$Ly = (D^2 + U)y = 0, \qquad D = \frac{r-2}{r}\partial_r, \qquad U = \omega^2 - \frac{6(r-2)(r-1)}{r^4}.$$

Elimination of the auxiliary component from the repeated spin-two extension gives

$$Lx = (s_1 D + s_0)y.$$

The scalar

$$\mathcal{I} = s_0 - \frac{1}{2}Ds_1$$

transforms under every rational triangular companion-frame change as

$$\mathcal{I} \longmapsto \mathcal{I} - \frac{1}{2}\mathcal{K}_U q, \qquad \mathcal{K}_U = D^3 + 4UD + 2D(U).$$

Hence it defines a projective class

$$[\mathcal{I}] \in \mathbb{Q}(i,\omega)(r)/\mathcal{K}_U\mathbb{Q}(i,\omega)(r).$$

For the action-derived extension this class is nonzero over the generic rational field. A declared pole reduction gives the representative

$$\boxed{\mathcal{I}_{\mathrm{red}} = \frac{i(r-2)(2r\omega^2 + 3\omega^2 + 12)}{5r^4\omega}.}$$

The local pole analysis makes the rational gauge ansatz exhaustive: after removal of the order-four apparent pole at $r = 2i/\omega$, only

$$\frac{c_{-2}}{r^2} + \frac{c_{-1}}{r} + c_0$$

can remain. The exact coefficient system has rank three and augmented rank four. Thus no rational gauge annihilates the class.

Moreover, with $f = (r-2)/r$,

$$[\mathcal{I}] \neq q(\omega)[-f/r^2]$$

for every generic rational q: the corresponding coefficient system has rank four and augmented rank five. The axial extension is therefore neither a physical Schwarzschild-parameter tangent nor the natural algebraic continuation in $\ell(\ell+1)$.

The left-null inconsistency witnesses also make the possible specialization locus finite. Rational splitting is excluded whenever

$$\omega \neq 0, \quad \omega \neq i, \quad \omega^2 \neq 3,$$

while angular proportionality is excluded whenever

$$\omega \neq 0, \quad \omega \neq i, \quad \omega^2 \neq -4.$$

No splitting or nonsplitting conclusion is inferred at the excluded or witness-zero specializations.

Equivalently, the repeated block defines an intrinsic class

$$e_{\mathrm{Bach}} \in \mathrm{Ext}^1_{\mathbb{Q}(i,\omega)(r)[D]}(M_{\mathrm{RW}}, M_{\mathrm{RW}}),$$

whose projective image is $[\mathcal{I}]$. The results above show that this generic class is not the jet of the frequency, Schwarzschild-mass, or angular-barrier families.

Proposition 9.9 (Intrinsic deformation jet and contour selector)**.** *Let $S = s_1 D + s_0$ be the scalar source of the repeated spin-two block and introduce the auxiliary family*

$$L_\tau = L + \tau S.$$

The equations $Ly = 0$ and $Lx + Sy = 0$ are exactly the first jet of $L_\tau(y + \tau x) = 0$ at $\tau = 0$. Over the dual numbers, multiplication by $a + \epsilon b$ in the ordered column basis $(\epsilon, 1)$ is

$$\begin{pmatrix} a & b \\ 0 & a \end{pmatrix}.$$

More generally, for

$$C(\tau) = \begin{pmatrix} a(\tau) & d(\tau) \\ 0 & f \end{pmatrix}, \qquad f'(0) = 0,$$

the partial jet in the ordered basis $(\epsilon\,\text{spin-2}, \text{spin-2}, \text{spin-1})$ is

$$J(C) = \begin{pmatrix} a & a_\tau & d_\tau \\ 0 & a & d \\ 0 & 0 & f \end{pmatrix}_{\tau=0}.$$

On this partially dualized triangular category,

$$J(C_1 C_2) = J(C_1) J(C_2), \qquad J(C^{-1}) = J(C)^{-1}.$$

The exact factor gauge realizes this structure locally. Use the state order

$$(x_0, x_1, z) = (\text{metric RW}, \text{carrier RW}, L_x).$$

Direct transformation of the complete six-state connection gives

$$\frac{d}{dr}\begin{pmatrix} x_0 \\ x_1 \\ z \end{pmatrix} = \begin{pmatrix} A & E & C \\ 0 & A & D \\ 0 & 0 & A_x \end{pmatrix}\begin{pmatrix} x_0 \\ x_1 \\ z \end{pmatrix}, \qquad E = USJ, \qquad C = USN,$$

where both E and the newly extracted mixed block C have rank one. In fact the statement is stronger:

$$[E\ C] = US[J\ N], \qquad \operatorname{rank}[E\ C] = 1.$$

Writing the rank-one matrix $US = \ell\rho$, the complete forcing of the upper spin-two row is the single scalar channel

$$\ell\,\sigma, \qquad \sigma = \rho(Jx_1 + Nz).$$

This is precisely the spin-two-row partial jet of

$$\mathcal{B}(\tau) = \begin{pmatrix} A + \tau E & D + \tau C \\ 0 & A_x \end{pmatrix},$$

with z held τ-independent. It is not the full jet of all four base states, which would be eight-dimensional. Accordingly, if the four-state fundamental map is

$$\Phi_4(\tau) = \begin{pmatrix} P(\tau) & Q(\tau) \\ 0 & R \end{pmatrix},$$

then every interior six-state fundamental map has the exact form

$$\Phi_6 = \begin{pmatrix} P & \dot{P} & \dot{Q} \\ 0 & P & Q \\ 0 & 0 & R \end{pmatrix}_{\tau=0}, \qquad \det \Phi_6 = (\det P)^2 \det R.$$

Consequently a typed identity $T_\pm = J(C_\pm)$ *would imply*

$$T_+ T_-^{-1} = J(C_+ C_-^{-1})$$

without reconstructing the cancellation between the jet coefficients after transport. This last reading requires both endpoint frames to be induced by the same analytic τ*-family; that compatibility and the resulting typed* T_+ *are not established here. An exact endpoint audit nevertheless closes the base-germ part of this requirement. After explicit scalar rescalings and the common permutation* $(R, S, E) \mapsto (E, R, S)$*, the metric column at the future horizon and at both null infinities is the* ϵ*-copy of the same scalar RW germ as the carrier base column; the spin-one normalization can be held* τ*-constant. The audit also recomputes the outgoing factor line. The endpoint audit itself leaves the differentiated recurrences and the upper-triangular shears* K_H, K_-, K_+ *open. A dedicated moving-phase successor answers the first horizon question. In the exact factor gauge,*

$$R_H = \begin{pmatrix} 0 & 0 \\ 3/2 & -1 - 4i\omega \end{pmatrix}, \qquad E_{-1} = 0,$$

so the perturbation of the selected regular exponent is

$$\dot{\lambda}_H = \frac{\ell_H^T E_{-1} r_H}{\ell_H^T r_H} = 0.$$

There is therefore no intrinsic $\tau \log(r-2)$ *term. Five coupled base/tangent Frobenius orders have been constructed in the shared* ω*-Taylor/dual-*τ *arithmetic. Exact Cauchy and pivot majorants bound the omitted base and tangent tails by* 3.5490×10^{-36} *and* 1.1768×10^{-34} *at* $\rho_0 = 2^{-22}$*. Across the first reduced pure-spin-two panel, the direct repeated-block and dual-number transports have equal Taylor coefficients and overlapping tail-enclosed output hulls. The spin-one Levelt initializer, mixed column, further panels, and hence the endpoint shear* K_H*, remain open. Thus, conditional on analytic endpoint-compatible scalar Jost frames, the off-diagonal incoming coefficient is*

$$b(\omega) = \partial_\tau a(\omega, \tau)\big|_{\tau=0}.$$

For a simple zero branch $a(\omega_n(\tau), \tau) = 0$*,*

$$\omega_n'(0) = -\frac{b(\omega_n)}{a'(\omega_n)} = -\frac{\beta_n}{\alpha_n}$$

under the compatible Fredholm normalization.

Let D_ω *be a spectral disk whose boundary is zero-free and contains distinct simple zeros* $\omega_1, \ldots, \omega_N$*. The contour moments*

$$K_k(D_\omega) = \frac{1}{2\pi i} \oint_{\partial D_\omega} \omega^k \frac{b(\omega)}{a(\omega)}\, d\omega$$

obey

$$K_k(D_\omega) = \sum_{j=1}^{N} \omega_j^k \frac{b(\omega_j)}{a'(\omega_j)}.$$

They are invariant under zero-free analytic endpoint renormalizations $a \mapsto ua$. In particular, if $N = 1$, then

$$K_0(D_\omega) = \frac{\beta_n}{\alpha_n};$$

a certified nonzero contour moment would therefore select the defective Smith branch without evaluating the unknown root itself.

For a cluster of N distinct simple zeros, the first N moments form a complete coordinate system for $\mathscr{A}_D = \mathcal{O}(D_\omega)/(a)$. If

$$P_D(z,\tau) = \prod_{j=1}^{N}(z - \omega_j(\tau)), \qquad Q_D(z) = \partial_\tau P_D(z,\tau)|_{\tau=0}\,,$$

then

$$Q_D(\omega_j) = \frac{b(\omega_j)}{a'(\omega_j)} P_D'(\omega_j).$$

Consequently $Q_D \not\equiv 0$ selects at least one defective QNM in the cluster, while $\mathrm{Res}(P_D, Q_D) \neq 0$ selects every QNM in the cluster as defective. Both polynomials can in principle be reconstructed from the ordinary argument-principle moments and the extension moments, without locating the individual roots.

The intrinsic organizing element is the QNM-velocity class

$$\boldsymbol{\kappa}_D = [b]\,[a']^{-1} \in \mathscr{A}_D.$$

The simple-root hypothesis makes $[a']$ a unit, and evaluation sends this class to

$$\left(\frac{b(\omega_1)}{a'(\omega_1)}, \ldots, \frac{b(\omega_N)}{a'(\omega_N)}\right).$$

Multiplication by $\boldsymbol{\kappa}_D$ therefore has rank equal to the number of defective QNMs in the cluster. In the monomial basis define

$$G = (S_{p+q}), \quad G_1 = (S_{p+q+1}), \quad H = (K_{p+q}), \quad H_1 = (K_{p+q+1}),$$

where $S_k = (2\pi i)^{-1} \oint \omega^k a'/a \, d\omega$. Then

$$M_\omega = G^{-1}G_1, \qquad M_\kappa = G^{-1}H$$

commute, obey

$$H_1 = G_1 G^{-1} H,$$

and have joint eigenpairs

$$\left(\omega_j, \frac{b(\omega_j)}{a'(\omega_j)}\right).$$

These are exact conditional cluster identities, not evaluated QNM data.

Equivalently, the Cauchy transform

$$R_D(z) = \frac{1}{2\pi i} \oint_{\partial D_\omega} \frac{b(\omega)}{(z-\omega)a(\omega)}\, d\omega = \sum_{j=1}^{N} \frac{\beta_j/\alpha_j}{z - \omega_j}$$

has poles precisely at the defective QNMs; semisimple QNMs are removable. Its reduced denominator is

$$\frac{P_D}{\gcd(P_D, Q_D)}.$$

If $H_N = (K_{p+q})_{p,q=0}^{N-1}$*, then*

$$H_N = V \ \mathrm{diag}\left(\frac{\beta_1}{\alpha_1}, \dots, \frac{\beta_N}{\alpha_N}\right) V^T,$$

where V *is the cluster Vandermonde matrix. Thus* rank H_N *is exactly the number of defective QNMs. A generic Loewner realization of* R_D *has the same rank, and its shifted pencil has precisely the defective QNM frequencies as finite generalized eigenvalues.*

Proposition 9.10 (Projective reduction of the intrinsic QNM shift). *Let* $Y_H(\omega, \tau)$ *and* $Y_\infty(\omega, \tau)$ *be analytic selected spin-two endpoint lines transported to a common interior point. In a projective chart valid on a spectral domain, write*

$$Y_\star = \gamma_\star \begin{pmatrix} 1 \\ v_\star \end{pmatrix}, \qquad \delta = v_\infty - v_H.$$

Then the scalar Evans coefficient factors as

$$a(\omega, \tau) = g(\omega, \tau)\delta(\omega, \tau), \qquad g = \gamma_H \gamma_\infty,$$

up to the fixed orientation sign. Whenever the chart is valid, g *is analytic and nonzero. At every simple QNM,*

$$\frac{\partial_\tau a}{\partial_\omega a} = \frac{\partial_\tau \delta}{\partial_\omega \delta}.$$

Moreover, on every contour contained in the chart domain,

$$\oint \omega^k \frac{\partial_\omega a}{a}\, d\omega = \oint \omega^k \frac{\partial_\omega \delta}{\delta}\, d\omega, \qquad \oint \omega^k \frac{\partial_\tau a}{a}\, d\omega = \oint \omega^k \frac{\partial_\tau \delta}{\delta}\, d\omega.$$

Thus the QNM count, the local velocity $\kappa_n = \delta_\tau/\delta_\omega$*, and all finite-cluster extension moments depend only on the two transported projective endpoint lines.*

For a companion system $Y_x = AY$*, the chart* $v = Y_2/Y_1$ *obeys*

$$v_x = A_{21} + (A_{22} - A_{11})v - A_{12}v^2,$$

while v_τ *and* v_ω *obey the corresponding linearized inhomogeneous equations. A fixed* $GL(2, \mathbb{C})$ *chart change acts by a Möbius transformation and preserves holomorphic parameter dependence. Consequently amplitude transport is unnecessary for the Evans/Smith selector; it is required only later for physically normalized* T_+ *and fluxes.*

The auxiliary τ is a bookkeeping coordinate on the intrinsic deformation class, not a physical parameter or a changed fundamental theory. The analytic complex-frequency Jost frames and the contour moment have not been constructed here. One exact contour prerequisite is now available: on the rational radius-0.025 seed disk, the horizon recurrence divisor

$$(n+1)(n+1+4i\omega)$$

and the infinity formal-recurrence divisor

$$2i\omega(n+1)$$

are uniformly nonzero, and all declared frame events and the real-exterior moving divisor $r\omega - 2i$ are excluded. The infinity series still lacks a direct recurrence remainder. A separate exterior-complex-scaled Volterra certificate now supplies a uniform scalar outgoing initializer on the same disk, but not yet the intrinsic-deformation tangent column, inward matching, an Evans boundary, or a root count. Likewise, differentiating a scalar flux identity requires an additional real/self-adjoint realization of L_τ, which is not supplied by the local algebra.

The operator $\mathcal{K}_U$ is the symmetric-square Regge–Wheeler operator: if $y_1, y_2 \in \ker L$, then

$$\mathcal{K}_U(y_1^2) = \mathcal{K}_U(y_1 y_2) = \mathcal{K}_U(y_2^2) = 0.$$

After an endpoint-admissible analytic gauge, the complete off-diagonal spin-two connection is consequently determined by the three periods

$$\int_\Gamma \mathcal{I} y_1^2 \, dr_*, \qquad \int_\Gamma \mathcal{I} y_1 y_2 \, dr_*, \qquad \int_\Gamma \mathcal{I} y_2^2 \, dr_*.$$

At a simple QNM the Smith selector is the rapid-decay period against y_n^2, together with the fixed endpoint-normalization term. Rational nontriviality does not force that particular period to be nonzero, so this proposition does not evaluate β_n, select a QNM Smith branch, or establish a double pole.

Theorem 9.11 (Exact future-horizon flux)**.** *For axial $\ell = 2$ and every real $\omega > 0$, the exact outward-oriented future-horizon Gram $H_{\mathcal{H}^+}$, evaluated from the correlated Frobenius series and the action current, has rank three and inertia*

$$\operatorname{inertia} H_{\mathcal{H}^+} = (1, 2, 0) \qquad (\alpha_{\mathrm{W}} > 0).$$

Its metric/carrier spin-two kernel plane has inertia $(1,1,0)$, while the orthogonal unit spin-one quotient has exact norm

$$-\pi \alpha_{\mathrm{W}} \frac{32}{15\omega}.$$

Neither $H_{\mathcal{H}^+}$ nor $-H_{\mathcal{H}^+}$ is positive semidefinite.

The all-positive-frequency extension follows from the symbolic recurrence: its residue eigenbasis and all factor-frame denominators are collision-free on $\omega > 0$, the three integer resonances have exact zero residual, and the omitted-head power count cannot change the Laurent constant.

The oriented Stokes identity is therefore

$$H_{\mathcal{H}^+} + T_+^\dagger G_+ T_+ - T_-^\dagger G_- T_- = 0.$$

The horizon theorem independently reproduces the hyperbolic-spin-two plus negative-spin-one anatomy of the incoming endpoint. Its indefiniteness also means that a semidefinite shortcut cannot determine $\operatorname{rank} T_+$.

Theorem 9.12 (Transport-free one-sided pseudo-isometry)**.** *On the pilot band the selected future-regular horizon family defines exact global ODE solutions. Unique expansion in the certified six-column infinity basis defines the typed maps T_- and T_+, without requiring their entries to be assembled at a common finite radius. Exact radial current conservation, the null trace-limit interchange, wrong-endpoint suppression, and the outward horizon orientation give*

$$H_{\mathcal{H}^+} + T_+^\dagger G_+ T_+ - T_-^\dagger G_- T_- = 0$$

on the complete future-regular three-space. Since T_- is invertible,

$$R = T_+ T_-^{-1}, \qquad A = T_-^{-1}, \qquad \mathsf{S} = \begin{pmatrix} R \\ A \end{pmatrix}$$

satisfy

$$\mathsf{S}^\dagger (G_+ \oplus H_{\mathcal{H}^+}) \mathsf{S} = G_-.$$

In particular, S *is injective because* A *is invertible.*

Normalize the incoming, outgoing, and horizon forms either to

$$J_\sigma = \operatorname{diag}(1, -1, -1) \quad \text{or} \quad J_0 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

The signature frame J_σ carries the invariant sign decomposition; the null frame J_0 carries factor triangularity. They must not be conflated. Sylvester congruences to either common form exist, although their explicit matrices have not been computed. In such normalized frames, with

$$R = \widehat{T}_+ \widehat{T}_-^{-1}, \qquad A = \widehat{T}_-^{-1}, \qquad \mathsf{S} = \begin{pmatrix} R \\ A \end{pmatrix},$$

one has

$$\mathsf{S}^\dagger (J \oplus J) \mathsf{S} = J, \qquad D := J - R^\dagger J R = A^\dagger J A.$$

Thus D is nondegenerate with inertia $(1, 2, 0)$, and S admits a noncanonical algebraic completion to $U(2, 4)$. The determinant is normalization-sensitive:

$$\det D = |\det \widehat{T}_-|^{-2} = |A_{\mathrm{in},2}|^{-4} |A_{\mathrm{in},1}|^{-2}.$$

Using the raw T_- without the endpoint normalizer ratio is incorrect. Equivalently, with the Krein adjoint $M^\sharp = J^{-1} M^\dagger J$, the identity is

$$R^\sharp R + A^\sharp A = I.$$

Because A is invertible, $I - R^\sharp R$ is invertible and hence $1 \notin \sigma(R^\sharp R)$. This does not exclude nonzero horizon waves of zero scalar flux: a nondegenerate indefinite horizon form still has a null cone.

Proposition 9.13 (Transport-free outgoing pullback test)**.** *In the raw horizon basis define*

$$\mathcal{O} := T_-^\dagger G_- T_- - H_{\mathcal{H}^+}.$$

Then

$$\mathcal{O} = T_+^\dagger G_+ T_+, \qquad \det \mathcal{O} = \det(G_+) |\det T_+|^2.$$

Since G_+ is nondegenerate,

$$T_+ \in GL(3, \mathbb{C}) \quad \Longleftrightarrow \quad \det \mathcal{O} \neq 0.$$

The zero/nonzero decision is invariant under every typed horizon and incoming basis change.

The following real-cell certificate decides this criterion without assembling either raw matrix.

Theorem 9.14 (Full outgoing population on a real-frequency cell)**.** *For Schwarzschild $M = 1$, axial $\ell = 2$, and every*

$$0.49995 \leq \omega \leq 0.50005,$$

validated scalar transport with one real Arb frequency ball gives the uniform bounds

$$|A_{\mathrm{out},2}| > 0.011326240435330133, \qquad |A_{\mathrm{out},1}| > 0.14680548488890086.$$

The exact endpoint-compatible RW/RW/spin-one filtration then implies

$$\ker T_+(\omega) = 0, \qquad T_+(\omega) \in GL(3, \mathbb{C})$$

throughout the declared cell. Consequently the complete three-dimensional outgoing trace space is populated on a nonzero real-frequency interval, and

$$\det \mathcal{O}(\omega) \neq 0, \qquad \operatorname{inertia} \mathcal{O}(\omega) = (1, 2, 0) \quad (\alpha_{\mathrm{W}} > 0).$$

Proof. If a future-horizon-regular datum belongs to $\ker T_+$, its outgoing spin-one quotient vanishes. The spin-one strict bound forces that scalar quotient solution to vanish, leaving the carrier spin-two subfiltration. Its outgoing quotient then vanishes, and the spin-two strict bound removes it. The remaining metric spin-two solution has the same nonzero scalar coefficient and therefore vanishes. Thus the kernel is zero. Since T_+ is 3×3, it is invertible. The conclusions for $\mathcal{O}$ follow from Proposition 9.13 and Sylvester congruence. □

The frequency dependence is enclosed directly in the coefficient balls and accumulated into the spatial defect at every Taylor step; the theorem is not an inference from finitely many sampled frequencies. In the intrinsic factor normalization the scalar diagonal product obeys uniformly

$$|A_{\mathrm{out},2}|^2 |A_{\mathrm{out},1}| > 0.00001883275407012474144974643389146.$$

No raw six-state determinant lower bound is inferred, because the endpoint frame units have not been numerically normalized. An older raw point calculation gives $\det \mathcal{O} \simeq 2.38117$, inertia $(1, 2, 0)$, and minimum absolute eigenvalue 0.0298; it remains explicitly classified as unvalidated numerical evidence and is not used in the theorem. An explicit T_+ remains useful for amplitudes and an independent end-to-end audit, but is no longer a prerequisite for the abstract pseudo-isometry or the outgoing-rank test. A three-parameter (α, γ, μ) reduction of the null-frame extension entries also remains open.

Corollary 9.15 (Generic and band-limited outgoing completeness)**.** *Assume the standard holomorphic dependence of the selected scalar Jost coefficients and typed factor connection on connected complex neighbourhoods of the positive real axis, with the threshold $\omega = 0$ excluded. Let*

$$Z_+ = \{\omega > 0 : A_{\mathrm{out},2}(\omega) A_{\mathrm{out},1}(\omega) = 0\}.$$

Then Z_+ is locally finite and

$$T_+(\omega) \in GL(3, \mathbb{C}) \qquad (\omega \in (0, \infty) \setminus Z_+).$$

Thus all three outgoing channels are populated on an open dense, full-measure subset of the positive real axis.

On $I_0 = [0.49995, 0.50005]$, multiplication by $T_+(\omega)$ is a bounded isomorphism of $L^2(I_0; \mathbb{C}^3)$. Hence every square-integrable outgoing packet supported in I_0 has a unique horizon-regular preimage. The pointwise Stokes identity integrates to

$$\int_{I_0} x^\dagger G_- x \, d\omega = \int_{I_0} (Rx)^\dagger G_+ (Rx) \, d\omega + \int_{I_0} (Ax)^\dagger H_{\mathcal{H}^+} (Ax) \, d\omega,$$

where $R = T_+ T_-^{-1}$ and $A = T_-^{-1}$.

On every compact $I \Subset (0, \infty)$, multiplication by T_+ is injective with dense range. It is a bounded isomorphism if $I \cap Z_+ = \varnothing$; if I contains an actual reflection zero, the range is dense and nonclosed.

Proof. The cell theorem shows that neither scalar outgoing coefficient is identically zero. Holomorphy makes each positive-real zero set locally finite, while the exact triangular determinant has only the diagonal zero divisor

$$\det T_+ = \rho_+ A_{\text{out},2}^2 A_{\text{out},1}, \qquad \rho_+ \neq 0.$$

Continuity and compactness then give a bounded inverse on I_0, and the integrated identity follows from the pointwise Stokes theorem.

For arbitrary compact I, T_+ is invertible almost everywhere. The multiplication operator and its adjoint consequently have zero kernel, so its range is dense. If I meets Z_+, continuity of the least singular value supplies unit vectors supported in shrinking neighbourhoods of a zero whose images tend to zero. The multiplier is not bounded below, and its dense range cannot be closed. $\square$

In the intrinsic factor normalization $\rho_+ = 1$, the cell certificate also gives

$$|\det T_+| = |A_{\text{out},2}|^2 |A_{\text{out},1}| > 1.8832754070124741 \times 10^{-5}.$$

In any other typed raw frame the right-hand side is multiplied by the corresponding nonzero endpoint unit. The diagonal inverse factors satisfy

$$|A_{\text{out},2}^{-1}| < 88.291, \qquad |A_{\text{out},1}^{-1}| < 6.812.$$

These are not bounds on the complete inverse: the uncomputed extension shears can still be large. On any simply connected zero-free complex neighbourhood of the cell one may instead write

$$T_+ = \operatorname{diag}(a, a, f) \begin{pmatrix} 1 & b/a & c/a \\ 0 & 1 & d/a \\ 0 & 0 & 1 \end{pmatrix}, \qquad a = A_{\text{out},2}, \quad f = A_{\text{out},1}.$$

The scalar diagonal therefore controls population and singularities, while the extension contributes a holomorphic unipotent shear.

At an isolated pure spin-two reflection zero with $m = \operatorname{ord} a$, $q = \operatorname{ord} b$, and $f \neq 0$, the full local Smith valuations are

$$\big(0, \min(m, q), 2m - \min(m, q)\big).$$

For a simple zero this is $(0, 0, 2)$ when $b(\omega_*) \neq 0$ and $(0, 1, 1)$ when $b(\omega_*) = 0$. At a pure spin-one zero of order n with $a \neq 0$, they are $(0, 0, n)$. At a simultaneous zero the complete classification is determined by

$$I_1 = (a, b, c, d, f), \qquad I_2 = (a^2, ad, bd - ac, af, bf), \qquad I_3 = (a^2 f).$$

This classifies a reflection zero if one is found; it does not assert that such a positive-real exceptional frequency exists.

For comparison, the Einstein problem admits a global finite-energy Schwarzschild scattering theory relating initial data and radiation spaces by Hilbert-space isomorphisms [23]. Flat-space conformal-higher-spin scattering also admits conformal-gravity states beyond ordinary Einstein gravitons [24]; it does not construct the Schwarzschild Bach connection considered here.

9.3 Polar mixed finite line and its exceptional wall

On the maximal restriction-stable polar module, choose the ordered formal basis (E, X_0, X_1, X_2). The denominator-cleared current matrices at powers $p = 0, -1, -2$ are anti-Hermitian. The $p = 0$ matrix has rank three and a one-dimensional mixed Einstein/additional radical. If y spans that line, then

$$y^\dagger N^{(-1)} y = 0, \tag{38}$$

while its first finite coefficient factors as

$$y^\dagger N^{(-2)} y = C\,\omega^{51}\Lambda^3(\Lambda-2)^5 P_6^2 P_{20} Q_{21}(\Lambda, \omega^2). \tag{39}$$

The constant C is nonzero, and the exact sum-of-squares and resultant certificates show that $P_6 P_{20} \neq 0$ for integer $\ell \geq 2$ and real $\omega \neq 0$. Therefore the filtered line is generically nonradical. At $Q_{21} = 0$ its deeper disposition remains open.

Writing $x = \widehat{\omega}^2 > 0$, the exact positive-root phase diagram is

Harmonic	positive x-roots	real nonzero $\widehat{\omega}$-roots
$\ell = 2$	0	0
$\ell = 3$	3	6
$4 \leq \ell \leq 10$	1	2
$11 \leq \ell \leq 40$	3	6
$\ell \geq 41$	1	2

At the legacy fixture $(\Lambda, x) = (6, 9/25)$,

$$Q_{21}(6, 9/25) = -\frac{174226120816040380076641138108451235935620694016}{227373675443232059478759765625} \neq 0. \tag{40}$$

The previously recorded longer rational was $Q_{21}(6, 81/625)$, caused by squaring x twice; it was neither the fixture value nor a Hilbert norm.

Theorem 9.16 (Generic polar formal radial disposition)**.** *Away from the explicit Q_{21} wall, a one-dimensional mixed Einstein/additional polar line is finite through the nonintegrable layers and has nonzero first finite pairing. Thus formal radial pairing finiteness does not canonically select a polar Einstein representative.*

This theorem neither extends the terminal-only polar prefixes nor matches the mixed line to future-horizon-regular data. In particular, the axial and polar infinity results do not identify a global solution admitted at both ends of the exterior.

9.4 Local Cauchy selection remains available

There is no no-go against local causal selection. On the globally hyperbolic exterior with Cauchy surface $\Sigma = \{t = \text{const}, r > 2m\}$, the local Cauchy condition

$$\psi|_\Sigma = 0, \qquad \nabla_n \psi|_\Sigma = 0 \tag{41}$$

propagates $\psi = 0$ in the axial sector unconditionally: the carrier operator equals $\frac{1}{2}\Box + C \circ (-)$ exactly there (the trace vanishes identically on the class), is normally hyperbolic, and its Bianchi constraint obeys the exact transport identity $\nabla^a (L\psi)_{a\varphi} = \frac{1}{2}\Box B_\varphi$. In the polar sector the operator is identically traceless at the PDE level, the conformal direction is uncontrolled, and $\psi_{\text{conf}}(t^4\chi(r)P_2)$

is an exact nonuniqueness witness with vanishing Cauchy data; the certified conformal trace relation $S(\psi_{\mathrm{conf}}(\Phi)) = -3\Box\Phi$ then reduces any zero-data solution to the conformal-gauge orbit, so local Cauchy data selects the Einstein image exactly *modulo conformal gauge* (and exactly on the traceless slice). Dropping the normal-derivative datum is certified to fail by a time-odd mode witness. The open physical questions are whether this restriction is preserved by nonlinear interactions and whether it follows from a differentiable asymptotic phase space rather than being imposed by hand.

10 Polar chain: composition, reach, current, and disposition

For polar $\ell = 2$ Regge–Wheeler-gauge perturbations, the trace variation δR need not vanish. The axial formula acquires two universal trace terms.

Corollary 10.1 (Polar trace-coupled composition)**.** *On the certified polar $\ell = 2$ carrier, Proposition 6.1 gives*

$$\boxed{\delta B_{ab} = \frac{1}{2}\Box(\delta R_{ab}) + C_{acbd}(\delta R)^{cd} - \frac{1}{6}\nabla_a\nabla_b\delta R - \frac{1}{12}g_{ab}\Box\delta R.} \tag{42}$$

The Einstein polar kernel injects by $\delta R_{ab} = 0$, while the curvature image obeys a second-order trace-coupled Lichnerowicz system for $(\delta R_{ab}, \delta R)$.

The polar chain is now certified through the same stations as the axial chain. Four structural facts organise it.

(i) Polar Einstein kernel. In the t-chart Regge–Wheeler polar gauge the traceless angular row of δR_{ab} equals $(H_0 - H_2)/2$, forcing $H_2 = H_0$; the remaining rows reduce exactly to a two-dimensional first-order system $dY/dr = M(r)Y$, $Y = (K, H_1)$, with H_0 algebraic. In horizon-adapted variables (K, BH_1) the ingoing residue spectrum is $\{0, -4im\omega\}$, matching the axial Regge–Wheeler control. A Schrödinger-form master scalar with ω-free potential was *not* found within the searched bounded rational ansatz classes; that anchor is recorded fail-closed open and nothing below depends on it.

(ii) Conformal gauge and polar reach. The polar carrier system on the Bianchi-constrained $(\delta R_{ab}, \delta R)$ is exactly three second-order equations in four radial functions: the operator (42) is identically traceless and divergence-free on the constraint surface, and the one-function underdeterminacy is linearized conformal gauge ($h = \Phi g$ is pure gauge and $\psi_{\mathrm{conf}} = \delta\mathrm{Ric}[\Phi g]$ annihilates every row for arbitrary Φ); this direction is absent in the axial sector. On the algebraic traceless slice the principal part diagonalises and the future horizon is a regular singular point with residue spectrum $\{0, 0, 0,\ 1 - 4im\omega,\ -1 - 4im\omega,\ -3 - 4im\omega\}$. Quotienting the regular conformal-gauge direction leaves a *two*-parameter physical ingoing-analytic polar carrier family at every $\omega \in \mathbb{R} \setminus \{0\}$ — the polar twin of the axial reach theorem.

(iii) Polar current. The sphere-integrated action-derived polar bilinear obeys the same off-shell 4α identity. On shell of the two-dimensional polar Einstein system every bilinear coefficient carries the factor $(\omega_1 + \omega_2)$: the polar Einstein self-block is exactly null for conjugate pairs, as in the axial sector. In addition the conformal direction Φg pairs to zero with an *arbitrary* off-shell polar perturbation: the sphere-integrated polar presymplectic form needs no conformal quotient at the bilinear level. Composition fixtures ($\omega = 3/5$, two interior radii, exact arithmetic) certify that the realized Ricci image closes on the full three-dimensional analytic carrier space — every ingoing-analytic polar carrier mode lifts to a metric perturbation with all seven $\delta\mathrm{Ric}$ rows verified — and that the Einstein–carrier cross pairing is nonzero. Cross-pairing nonvanishing is representative-independent because the Einstein block is null; the positive carrier self-norms are certified at the canonical composed representatives only, and the representative-invariant sign theory is open.

(iv) Polar disposition. The polar carrier's asymptotic symbol is $(\lambda^2 - \omega^2)^3$: it propagates on the Einstein characteristics, with Coulomb log-phases and amplitude falloffs r^{-1}, r^{-2}, r^{-3} — all decaying at real frequencies, strictly stronger than the axial r^0, r^{-1} case. The tested endpoint diagnostics (future-horizon analyticity, leading outer symbol and falloff class) therefore fail to separate the polar Ricci carrier from the polar Einstein carrier, exactly as in the axial sector: the endpoint-nonselection result is now *parity complete* at the $\ell = 2$ linear mode level. As before, this is not a classification of all local differential boundary operators.

(v) Polar formal infinity module. A later full-seven-row audit found that the shallow four-state master used in the earlier norm-selection table contained a source-zero direction that did not satisfy the complete Ricci system. The restriction-stable replacement and its current filtration are summarized by Equation (39): a mixed Einstein/additional line is finite and generically nonradical. The former power-enhanced/logarithmic fixture and parity-complete Einstein-only norm-selection theorem are therefore superseded. The independently verified polar composition, horizon reach, Einstein isotropy, conformal degeneracy, and controlled horizon pairings in items (i)–(iv) are unaffected.

11 Horizon monodromy temperature

One statement of Hawking type is available at the REDUCED-MODE level without any Lorentzian quantum construction. The surface gravity of the certified background is $\kappa = 1/(4m)$ with $T_H = \kappa/2\pi = 1/(8\pi m)$; the geometric-clock value agrees exactly with the certified normalized-frame first-law temperature.

Theorem 11.1 (Monodromy universality)**.** *Under the horizon continuation $\rho \to e^{2\pi i}\rho$, every certified ingoing residue exponent of the four $\ell = 2$ mode families (axial and polar, Einstein and Ricci carrier) has continuation factor exactly 1 or $e^{8\pi m\omega} = e^{\omega/T_H}$, and every family contains exponents with the nontrivial factor.*

The integer parts of the exponents are monodromy-trivial; the universal $-4im\omega$ part carries the thermal factor. Hence the Boltzmann ratio $|\beta/\alpha|^2 = e^{-\omega/T_H}$ of the Damour–Ruffini continuation is identical for the Einstein and additional curvature families in both parity sectors: at the mode level, the additional carrier is thermally weighted at exactly the Hawking temperature, and by the certified nonzero mixed pairings it participates in the horizon flux with that weight. This is a mode-level (REDUCED-MODE) statement only: no Hadamard state, renormalized stress tensor, grey-body factor, luminosity, back-reaction, or LORENTZIAN-CAUSAL Hawking theorem is claimed, and none exists in the repository.

12 Result ledger

Question	Certified answer	Scope
Static Bach-flat black holes?	exact Laurent-family classification and three-horizon fixture; Einstein sheet corrected	local/static
Differentiable energy?	residual-basic normalized generator, exact static first law	$u \neq 0$; signed orientation; $f(J)$ freedom
Einstein axial waves present?	yes; exact Regge–Wheeler injection	Schwarzschild, axial $\ell = 2$
Additional axial content?	nonzero second-order Ricci carrier; no canonical metric split	same
Reach future horizon?	two analytic ingoing ψ amplitudes	nonzero-real-frequency carrier
Carry current?	Einstein self-block exactly null; mixed/additional blocks nonzero in fixtures	exact identity plus three frequencies
Selected by tested endpoints?	no: horizon and leading-symbol tests do not force $\psi = 0$	not a general causal no-go
Formal radial infinity?	On the complete axial $\ell = 2$ pilot, repaired X_0 is finite in the fixed representative and under rate-zero Einstein shears, but oscillatory Einstein shears cross it divergently; the polar sector has a generic-angular mixed finite line	no representative-invariant axial quotient, phase space, norm, or global matching
Null-endpoint wave-packet flux?	Exact three-dimensional axial L^2 trace spaces at both null ends; Grams have rank three, zero radical, and inertia $(1, 2, 0)$ for $\alpha_{\mathrm{W}} > 0$	$\ell = 2$, $M = 1$, $\omega \in [1/2, 3/4]$
Incoming global channels?	Exact RW/RW/spin-one triangular filtration and $T_{-}(\omega) \in GL(3, \mathbb{C})$ for every real $\omega > 0$; the full incoming indefinite trace is populated; on $0.49995 \leq \omega \leq 0.50005$, $T_{+} \in GL(3, \mathbb{C})$ and the full outgoing trace is populated; analyticity gives outgoing population off a locally finite positive-real exceptional set and dense range on every compact positive-frequency band	outgoing amplitudes, absence or location of isolated reflection zeros, complete-pilot pointwise T_{+} rank, non-real damped poles, time-domain stability, CPT and complete scattering open
Polar exceptional wall?	generically nonradical finite line; exact wall $Q_{21}(\ell(\ell+1), \widehat{\omega}^2) = 0$ with certified harmonic root counts	deeper filtration open on the wall
Einstein–extra horizon pairing invariant?	Einstein block isotropic; nonzero cross pairing and full rank in controlled fixtures; extra self-pairing sign not invariant	axial $\ell = 2$, real $\omega \neq 0$
Complex-frequency structure?	the complete non-split axial $\ell = 2$ system has no lower-half-plane growing separated mode in the $e^{+i\omega t}$ convention; the selected frames have damped-half-plane events at $\{i/4, i/2, i\}$	time-domain PDE stability, regularized damped poles, and the quasinormal spectrum remain open
Static ($\omega = 0$) sectors?	log-classified: two-parameter log-free families both parities; polar Einstein variables degenerate	classification only; static composition open
Polar composition?	exact trace-coupled identity; realized image closes on the full analytic carrier space	Schwarzschild, polar $\ell = 2$
Polar reach and current?	two physical ingoing amplitudes modulo conformal gauge; Einstein self-block null; nonzero mixed fixtures	conformal direction an exact off-shell degeneracy
Polar disposition?	Einstein characteristics, all-decaying Coulomb asymptotics; endpoint nonselection parity complete	real frequencies, $\ell = 2$

13 What is not proved

The paper does not establish:

- a complete exterior initial-boundary well-posedness theorem: the real-frequency incoming connection is now exact and invertible on the entire positive real axis, and a real cell around $\omega = 1/2$ has full outgoing population and a bounded L^2 multiplier inverse; analyticity gives generic positive-real outgoing population off a locally finite exceptional set, but explicit outgoing amplitudes, pointwise classification at the exceptional frequencies and the complex-frequency connection remain unclassified;
- convergence or summability of the generic-ℓ formal series, axial X_2, the deeper polar filtration on the $Q_{21} = 0$ wall, the terminal-only polar prefixes, or a non-Einstein-background Ricci-carrier composition;
- a canonical splitting of the Bach solution space, or complete metric reconstruction from every Ricci-carrier solution;
- a representative-invariant sign theory for the carrier self-pairing (the null-quotient pairing);
- time-domain boundedness, decay, completeness, quasinormal modes, or ringdown waveforms: boundary dévissage excludes growing separated modes of the complete non-split axial $\ell = 2$ system, but does not prove a full PDE stability theorem; the damped-half-plane frame events $\{i/4, i/2, i\}$ are not classified as Evans zeros or nonzeros;
- a second-order physical-process first law or nonlinear horizon dynamics;
- asymptotically flat conformal-gravity scattering with a complete boundary phase space, metric/tetrad falloffs, gauge-audited charge algebra, and horizon-to-infinity connection map: the scoped endpoint L^2 trace spaces and Grams do not by themselves supply these global structures;
- Hawking states, particles, grey-body factors, luminosity, a quantum BRST quotient, positivity, or unitarity: the monodromy temperature is a mode-level continuation factor, not a quantum state construction.

In particular, the paper withdraws the broad phrase "no local causal truncation." Zero Cauchy data for the Ricci carrier is a local linear truncation. What is established is failure of the tested endpoint diagnostics to imply that truncation.

14 Certificate and reproduction ledger

The manuscript is an OUTLINE-ALLOWED/DRAFT-ALLOWED integration of committed certificates at repository baseline recorded in its generated claim map.

Claim	Authoritative certificate
Static spherical family	`black_hole_programme/certificates/BH0_STATIC_SPHERICAL_BACKGROUND.json`
Normalized generator and first law	`black_hole_programme/certificates/BH1A_NORMALIZED_GENERATOR.json`
Linear spherical dynamical extension	`black_hole_programme/certificates/BH1B_DYNAMICAL_EXTENSION.json`

Axial operator and Ricci–Bach composition	`black_hole_programme/certificates/BH2A_AXIAL_OPERATOR.json`
Extra horizon reach	`black_hole_programme/certificates/BH2A_HORIZON_REACH.json`
Action-derived axial current	`black_hole_programme/certificates/BH2A_FLUX_MATRIX.json`
Mixed and extra horizon flux fixtures	`black_hole_programme/certificates/BH2A_CROSS_FLUX.json`
Axial causal disposition	`black_hole_programme/certificates/BH2A_CAUSAL_DISPOSITION.json`
Polar trace-coupled split	`black_hole_programme/certificates/BH2B_POLAR_SPLIT.json`
Polar carrier horizon reach (conformal gauge quotient)	`black_hole_programme/certificates/BH2B_POLAR_REACH.json`
Polar Einstein two-dimensional reduction	`black_hole_programme/certificates/BH2B_POLAR_EINSTEIN.json`
Polar Einstein self-block null; conformal degeneracy	`black_hole_programme/certificates/BH2B_POLAR_FLUX.json`
Polar composition and mixed flux fixtures	`black_hole_programme/certificates/BH2B_POLAR_CROSS_FLUX.json`
Polar causal disposition	`black_hole_programme/certificates/BH2B_POLAR_DISPOSITION.json`
Horizon monodromy temperature	`black_hole_programme/certificates/BH4_HAWKING_MONODROMY.json`
Static-sector log classification	`black_hole_programme/certificates/BH2_OMEGA_ZERO.json`
Generic-ℓ axial Ricci-carrier recurrences (complete metric and current dispositions superseded)	`black_hole_programme/phase2/general_l_axial_asymptotics/certificate.json`
Complete axial six-dimensional module, repaired X_0, and legacy E_0 supersession	`black_hole_programme/phase3/axial_complete_reconstruction_repair/certificate.json`; `reports/phase3-black-hole-axial-complete-reconstruction-repair-2026-07-22.md`
Exact axial null-endpoint wavepacket traces and action-derived flux Grams	`black_hole_programme/phase3/axial_wavepacket_null_trace/certificate.json`; `black_hole_programme/phase3/axial_null_flux_gram/certificate.json`; `reports/phase3-black-hole-axial-null-flux-gram-2026-07-23.md`

Axial RW/RW/spin-one filtration and invertible incoming connection	black_hole_programme/phase3/axial_rw_lx_triangular_preflight/certificate.json; black_hole_programme/phase3/axial_incoming_connection_analytic/certificate.json; black_hole_programme/phase3/axial_incoming_extended_domain_audit/certificate.json; reports/phase3-axial-incoming-extended-domain-audit-2026-07-23.md; black_hole_programme/phase3/axial_boundary_devissage_no_growth/certificate.json; reports/phase3-axial-boundary-devissage-no-growth-2026-07-23.md; black_hole_programme/phase3/axial_horizon_grassmann_mobius_to_r4_taylor2/future_horizon_outward_gram.json; black_hole_programme/phase3/axial_horizon_grassmann_mobius_to_r4_taylor2/future_horizon_factor_quotient.json; black_hole_programme/phase3/axial_spin_two_scattering_extension_preflight/certificate.json; black_hole_programme/phase3/axial_qnm_local_smith_dichotomy/certificate.json; black_hole_programme/phase3/axial_qnm_parameter_deformation_cohomology_v1/certificate.json; black_hole_programme/phase3/axial_qnm_projective_cocycle_v1/certificate.json; black_hole_programme/phase3/axial_qnm_intrinsic_deformation_jet_v1/certificate.json; black_hole_programme/phase3/axial_qnm_endpoint_germ_divisor_v1/certificate.json; black_hole_programme/phase3/axial_qnm_infinity_tail_gate_v1/certificate.json; black_hole_programme/phase3/axial_qnm_ecs_inverse_tortoise_v1/certificate.json; black_hole_programme/phase3/axial_qnm_ecs_inward_transport_v1/certificate.json; black_hole_programme/phase3/axial_qnm_ecs_tangent_initializer_v1/certificate.json; black_hole_programme/phase3/axial_qnm_ecs_panel_transport_preflight_v1/certificate.json; black_hole_programme/phase3/axial_qnm_ecs_riccati_transport_preflight_v1/certificate.json; black_hole_programme/phase3/axial_qnm_projective_sensitivity_v1/certificate.json; black_hole_programme/phase3/axial_qnm_projective_evans_riccati_rail_v1/certificate.json; black_hole_programme/phase3/axial_qnm_projective_evans_riccati_rail_v2/certificate.json; black_hole_programme/phase3/axial_qnm_projective_evans_riccati_rail_v3/certificate.json; black_hole_programme/phase3/axial_qnm_ecs_centered_projective_initializer_v1/certificate.json; black_hole_programme/phase3/axial_qnm_ecs_affine_projective_transport_v1/certificate.json; black_hole_programme/phase3/axial_qnm_horizon_projective_preflight_v1/certificate.json; black_hole_programme/phase3/axial_qnm_horizon_reciprocal_chart_transport_v1/certificate.json; black_hole_programme/phase3/axial_qnm_horizon_reciprocal_checkpoint_transport_v1/certificate.json; black_hole_programme/phase3/axial_qnm_two_sided_projective_mismatch_preflight_v1/certificate.json; black_hole_programme/phase3/axial_qnm_common_affine_export_contract_v1/certificate.json; black_hole_programme/phase3/axial_qnm_common_affine_evans_boundary_v1/certificate.json; black_hole_programme/phase3/axial_partial_jet_transport_crosswalk_v1/certificate.json; black_hole_programme/phase3/axial_partial_jet_endpoint_frames_v1/certificate.json; black_hole_programme/phase3/axial_partial_jet_transport_preflight_v1/certificate.json; black_hole_programme/phase3/axial_partial_jet_horizon_moving_phase_v1/certificate.json; black_hole_programme/phase3/axial_partial_jet_horizon_spin_one_levelt_v1/certificate.json; black_hole_programme/phase3/axial_partial_jet_horizon_multipanel_preflight_v1/certificate.json; black_hole_programme/phase3/axial_partial_jet_horizon_projective_multipanel_v1/certificate.json; black_hole_programme/phase3/axial_partial_jet_horizon_checkpoint_frame_v1/certificate.json; black_hole_programme/phase3/axial_partial_jet_horizon_checkpoint_frame_v1/pivot-switch-certificate.json; black_hole_programme/phase3/axial_partial_jet_horizon_checkpoint_frame_v1/pivot-switch-continuation-certificate.json; black_hole_programme/phase3/axial_partial_jet_horizon_checkpoint_frame_v1/pivot-switch-subdivision-retry-certificate.json; black_hole_programme/phase3/axial_partial_jet_horizon_checkpoint_frame_v1/pivot-switch-shared-remainder-preflight-certificate.json; black_hole_programme/phase3/axial_partial_jet_horizon_checkpoint_frame_v1/shared-remainder-multipanel-successor-certificate.json; black_hole_programme/phase3/axial_partial_jet_horizon_checkpoint_frame_v1/adaptive-chart-separation-certificate.json; black_

Polar restriction-stable module and invariant current filtration	black_hole_programme/phase2/general_l_polar_extendible_current_closure/certificate.json
Generic-ℓ two-parity formal-infinity phase diagram	black_hole_programme/phase2/generic_l_synthesis/certificate.json
Local Cauchy truncation selects Einstein modulo conformal gauge	black_hole_programme/certificates/BH_LOCAL_EINSTEIN_CAUCHY_TRUNCATION.json
Horizon cross invariant and complex-frequency continuation	black_hole_programme/certificates/BH2_SYMBOLIC_CROSS_INVARIANT.json; black_hole_programme/certificates/BH3_ANALYTIC_CONTINUATION_GATE.json

Every certificate records a schema, exact or controlled-numerical payload, independent verifier, mutations or controls, command receipt, dependency tags, and missing claims. The former axial and polar infinity-flux certificates, the infinity-selection half of the endpoint assembly, and the additional-log-tail half of the exterior boundary-value gate are historical inputs, not active authorities for infinity selection. Their unaffected Einstein, horizon, and leading-symbol components are represented above by independent certificates. The exact superseded identifiers are recorded in the machine-readable claim map. Exact axial endpoint trace spaces and flux Grams now combine with the invertible incoming connection to give globally populated incoming channels. A complete asymptotically flat scattering theorem still requires the outgoing/horizon map, regular-tetrad falloff, differentiable phase space, charge algebra, and complex-frequency audit.

15 Outlook

The formal infinity problem now has a different shape from the one suggested by the original fixture. The complete axial $\ell = 2$ pilot contains a finite repaired non-Einstein direction, but that finite class is stable only under the rate-zero Einstein shear and not under the true oscillatory shear. The polar module separately contains a generic-angular, generically nonradical mixed finite line, with a finite algebraic set of exceptional frequencies at each harmonic. These facts neither establish a representative-independent axial quotient nor restore an Einstein-selection principle.

The first global map is now exact:

$$\text{future-horizon-regular data} \xrightarrow[\cong]{T_-} \text{incoming } \mathscr{I}^- \text{ trace.}$$

The remaining decisive map is the outgoing/horizon scattering relation. It must include a regular tetrad or gauge, a gauge-audited Lee–Wald charge algebra, and the connection amplitudes at both future boundaries. A first validated full-amplitude attempt certified its diagonal carrier and kernel ranks and all local lower-lift solves on the first frequency cell, but ended in SHORTFALL: fixed interval frames discarded the shared ω-dependence needed for cumulative lower-block cancellation. The analytic triangular proof bypasses that limitation for T_-, not for T_+. The partial-jet identity above now gives a representation-level successor: propagate the base spin-two/spin-one connection and its intrinsic dual-number tangent in one correlated algebra, so that inversion and frame composition retain the $bd - a_2c$ cancellation exactly. This is an implementation route, not yet a recovered outgoing map; the endpoint recurrences and frame normalizers must first be shown to arise from the same analytic jet family. Moreover the dual-number coefficient ring must sit over a shared validated ω-Taylor model. Dual numbers over independent interval boxes would retain the τ-jet but still lose the frequency correlations that caused the original refusal. This construction does not repair the separate H4 exterior-norm witness, whose shared-frequency rail failed for a different sum-of-squares conditioning reason. The first mixed-arithmetic rail $\mathrm{Taylor}_4(\omega) \otimes \mathbb{Q}[\epsilon]/(\epsilon^2)$ has now been exercised on the first near-horizon panel. It refuses at its analytic-tail gate: the direct majorant is noncontractive. This is a certified method shortfall, not a failure of the partial-jet identity or of the physical transport. A useful successor must factor the endpoint singular structure before applying the Taylor tail bound. The phase-factored horizon successor confirms this diagnosis: $\dot{\lambda}_H = 0$, so no new tangent logarithm occurs, and a tail-enclosed reduced Frobenius seed and first pure-spin-two panel pass. The spin-one companion block requires its own Levelt/phase reduction before a three-channel initializer and mixed column are bounded. That reduction is now exact:

$$\widehat{Z} = \mathrm{diag}(\rho, \rho^2)Z, \qquad \widehat{X} = \rho X, \qquad \widehat{Y} = \rho Y.$$

The order-one resonance is compatible with no forced logarithm; its omitted base and tangent tails are below 2.736×10^{-32} and 6.867×10^{-31}. The complete mixed first panel passes, including agreement of the direct regular six-state and partial-jet routes. The two routes continue to overlap for 27 geometric panels. They then refuse on cumulative rectangular width 1.164×10^{6}, while the local order-24 operator tail is only 5.56×10^{-27}. Thus the remaining horizon obstruction is correlation-preserving multipanel reconditioning, not endpoint singularity or local truncation. A fixed projective pivot improves the local scaling and completes five finer panels, but its generic complex Krawczyk inversion then refuses because the rectangular enclosure of the pivot norm spans zero; the local scaled norm is only 0.08741 and the operator tail 2.81×10^{-24}, so this is again a dependency failure rather than evidence of a physical pivot zero. The permitted midpoint/Lohner alternate exceeded a bounded 240-second runtime before flushing a panel checkpoint. It is recorded as a throughput shortfall, not as a transport obstruction. The endpoint normalization question can, however, be separated from that radial transport. In the canonical τ-analytic Frobenius/Levelt frame, the spin-two leading pairing and spin-one quotient leading coefficient are fixed independently of τ, and the sole compatible order-one Levelt free coefficient is set to zero for the complete analytic family. The exact recurrence audit then gives

$$K_H^{\mathrm{canonical}} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

This is the endpoint jet-frame shear, not the future-horizon Gram and not an $r = 4$ horizon frame. A new correlation-preserving checkpoint rail accepts 27 panels at $\omega = 4097/8192$; it then refuses because the mixed spin-one projective pivot enclosure contains zero at $\rho = 91/268435456$, while the pure spin-two pivot still has certified modulus lower bound 0.839407. Thus the canonical endpoint shear is fixed, but the complete three-column H_4 handoff remains open. A pre-emptive chart repair now switches at the last nondegenerate checkpoint using the fixed row $e_2 - e_3$. Its modulus lower bound is 1.35225; the exact dual normalization restores base pivot 1 and tangent pivot 0. One post-switch panel passes with scaled norm 0.39018, local tail 8.36×10^{-8}, and next-pivot lower bound 0.920325. This supplies a resumable correlation-preserving checkpoint beyond the former obstruction. A subsequent audit found, however, that the first continuation rail tested finiteness before rather than after projective normalization. Its advertised panel 31 is therefore superseded: the raw Taylor state and pivot pass, but normalization creates non-finite tangent balls. A mutation test reproduces the old false acceptance and the corrected gate rejects it. The last certified checkpoint is panel 30 at $\rho = 95/268435456$. A bounded retry grid comprising Taylor orders $16, 20, 24, 32, 40$ and subdivisions $2, 4, 8, 16, 32, 64$ completes none of the 30 attempted next panels: 20 stop at non-finite projective normalization and 10 because the sole admissible e_2 pivot contains zero. Thus the endpoint switch itself remains certified, but its radial continuation beyond panel 30 requires a correlation-preserving representation stronger than rectangular Taylor balls. A bounded shared-remainder repair now supplies exactly that representation for the next step. It computes one certified reciprocal of the projective pivot and reuses it in every normalized base and tangent coordinate, rather than forming a dependency-destroying squared-denominator enclosure. Starting from the corrected panel-30 checkpoint, all post-normalization state and amplitude balls remain finite and panel 31 at $\rho = 3/8388608$ is certified. A mutation test confirms that the eager squared-denominator implementation contains zero and fails. This is one positive repaired panel, not yet a continuation to $r = 4$. Subdividing the same shared-reciprocal rail by 128 certifies nine more normalized substeps, reaching

$$\rho = \frac{12297}{34359738368}.$$

The Taylor gate remains healthy at the terminal attempt, with scaled norm 0.002761 and finite tail 0.01921, but every member of the fixed projective atlas then has zero modulus lower bound. A midpoint-derived Möbius row also fails on the full ball. In fact each Cartesian base component contains zero, so the product enclosure contains the zero vector; no fixed linear chart can certify a nonzero denominator from this representation. The checkpoints cannot be upgraded after the fact: frequency was specialized before transport, seed coordinates were inflated independently, and the radial Taylor coefficients were discarded. A correlation-preserving rerun must therefore restart at the symbolic mixed Levelt seed $\rho = 2^{-22}$, before fixed-frequency substitution and initial normalization, and serialize a shared ω-Taylor/dual-τ affine state. This is a representation obstruction, not evidence that the true horizon line contains zero. The required restart has now passed its first constructive gate. On the disk

$$\left|\omega - \frac{4097}{8192}\right| \leq 2^{-22},$$

a degree-four seed carries one shared complex-frequency generator through both base and dual-τ rails, uses one coupled ℓ^∞ residual rather than independent component boxes, and certifies one radial successor. The initial and successor pivot lower bounds are 0.999874 and 0.997894; the residual grows from 0.00130228 to 0.00553187, while the radial Cauchy scaled norm is 0.480079 and its tail 0.000626096. This certifies the correlation format and one content-addressed step only. A multipanel continuation and the horizon H_4 gate remain open. A first 33-step continuation attempt was terminated after 344 seconds without producing an artifact and is recorded as a throughput shortfall, not a pass. The replacement contract is to serialize and independently verify the joint derivative kernel once, then emit one durable content-addressed radial child per invocation bounded by 60 seconds. Two subsequent 58-second attempts to serialize the complete joint derivative kernel also time out, including a unique-expression cache variant. They emit neither a kernel nor a radial child. The exact resume state remains $\rho = 65/268435456$ with model hash `48683b91...1f250`; the next admissible implementation must shard the kernel by generator entry and derivative order, hash-assemble those shards, and only then attempt one radial child.

At outgoing real infinity the corresponding reduced-phase representation has also passed its first local test. Keeping $e^{-2i\omega r_*}$ symbolic, the exact R_+ factor head has base and tangent residual valuations 6 and 4. The exact RW potential integral from $r = 32$ is 189/1024, giving the uniform Volterra contraction $189/512 < 1$ on the first frequency child. The resulting all-order shared-ω, τ seed remainder is attached before transport, and the macropanel $32 \to 1023/32$ passes with identical direct and dual-number Taylor coefficients. This certifies one repeated-spin-two outgoing Jost column. A checkpointed successor carries the same correlated column through 16 further panels to $r = 63/2$, again with exact direct/dual coefficient agreement; its final width is 1.61820 and its largest local tail is 1.951×10^{-9}. The abandoned monolithic 896-panel attempt exceeded the scoped runtime budget and is recorded only as a throughput failure. Continuation must resume from a serialized correlated checkpoint rather than replay from $r = 32$. The complementary columns, K_+, and T_+ remain open. The required restart format is now certified: it serializes the full mixed base/tangent

$\mathrm{Taylor}_4(\omega)$ state at $r = 63/2$, including exact coefficients, IEEE-754 interval endpoints, the shared frequency generator and content hashes. Its loader round-trips bit exactly, and a replay-free 16-panel continuation reaches $r = 31$ and agrees exactly with an independent 32-panel reference. The timing also isolates the implementation bottleneck: about 60 seconds are spent in Python symbolic preparation, while native 32-panel transport takes 9.00 seconds and the restarted 16-panel chunk 5.66 seconds. Checkpoint reuse, rather than a different ODE integrator, is therefore the immediate performance repair. The outgoing endpoint algebra is now sharper. In the typed factor order (E, R, S), the exact formal frame is

$$\left(2\,\mathrm{EI2},\ \mathrm{XI2} - \frac{i(16\omega^2 - 4i\omega - 5)}{\omega}\mathrm{XI3},\ \frac{i}{2\omega}\mathrm{XI3}\right),$$

with $\pi_x(R) = 0$ and $\pi_x(S) = 1$. The E line is the unit ϵ-copy of the already certified R scalar Jost column. The log-free formal recurrences, with the free EI2 coefficient fixed to zero throughout the family, give $K_+^{\mathrm{formal}} = 0$. This is not yet an analytic endpoint-frame theorem. The existing all-order six-state enclosure for XI3 originally rectangularized its base and tangent data and exported neither the common ω generator nor the dual-τ remainder required for the normalized S column. A new endpoint adapter now applies the exact factor transform $S = i\,\mathrm{XI3}/(2\omega)$, preserves generator 7315 and the degree-four head, and encloses the transformed all-order hull in one partial dual-τ remainder. This closes the representation contract at $r = 32$, but the tangent width is 369.996, against base width 1.35×10^{-3}. No inward S transport or jointly propagated (E, R, S) frame follows from that endpoint result alone. After exact internal tangent normalization by 512, however, a first inward panel of width 1/256 is now certified:

$$r : 32 \longrightarrow 8191/256.$$

The direct and partial-dual Taylor coefficients agree exactly, the interval difference contains zero, and the rigorous local tail is 7.72×10^{-10}. The normalized tangent width changes only from 0.722649 to 0.723408, although restoring the physical normalization gives width 370.385. This is a valid correlated checkpoint, not yet a jointly propagated frame. A replay-free successor now advances the same normalized S_+ state through 16 further panels,

$$r : 8191/256 \longrightarrow 8175/256.$$

All generator and direct/dual correlation gates pass; the largest local tail is 7.68×10^{-10}, the final base width is 0.00135325, and the normalized tangent width is 0.737306 (physical width 377.501 after restoring the factor 512). A second checkpointed successor adds 32 panels,

$$r : 8175/256 \longrightarrow 8143/256,$$

with generator 7315 and all direct/dual gates preserved. Its largest tail is 7.10×10^{-10}, final base width 0.00147618, and normalized tangent width 0.775139 (physical width 396.871). Consequently analytic K_+, the joint (E, R, S) frame, and T_+ remain fail closed. A one-step width ladder at this checkpoint distinguishes mathematical finiteness from an operational enclosure: width 1/128 has tail 5.53×10^{-6}, base width 0.001497, and normalized tangent width 0.7780, while widths 1/64 and 1/32 remain finite but fail the declared tail/base thresholds. Using the admissible width, a content-addressed replay-free continuation advances through 103 panels of width 1/128 and one final panel of width 1/256, reaching

$$r = 31.$$

All panelwise generator and direct/dual correlation gates pass. At the common radius with the existing R_+ checkpoint, the S_+ base width is 0.006021, the normalized tangent width is 1.564, and the restored physical tangent width is 800.528. The typed assembly has now been checked. In complex factor order (X, Y, Z),

$$E = (R_{\text{base}}, 0, 0), \qquad R = (R_\tau, R_{\text{base}}, 0), \qquad S = (S_{\tau,Y}, S_Y, S_Z).$$

The minor on rows (X_0, Y_0, Z_0) is $R_{\text{base},0}^2 S_{Z,0}$. Exact hulls give

$$\Re R_{\text{base},0} \in [1.00307, 1.00508], \qquad \Re S_{Z,0} \in [3.31 \times 10^{-5}, 2.209 \times 10^{-3}],$$

so the reduced outgoing $E/R/S$ frame has complex rank three throughout the frequency child. The R/E and S artifacts still use different nonzero analytic phase gauges, however. Formal canonical $K_+ = 0$ is preserved, but its analytic endpoint-τ promotion and T_+ remain unproved. The exact moving-phase audit explains why a static rephasing is insufficient. For the outgoing spin-two pencil,

$$\dot{q}_+ = -\frac{3}{4}, \qquad \dot{p}_+ = 0,$$

and the irregular coefficient is removed by

$$B = \operatorname{diag}\left(\frac{3}{4}, 0\right), \qquad E_{-1} + [A_0, B] = 0.$$

The combined scalar and polynomial-eigenvector generator is therefore $\operatorname{diag}(0, -3/4)$, whereas the spin-one phase is τ-independent. At $r = 31$, a common moving gauge requires the exact tangent correction

$$\operatorname{diag}\left(0, \frac{93}{4}\right).$$

The checkpoint has now been reissued with this correction on the R and S spin-two rows, while retaining the shared generator 7315 and the entire zero-free scalar gauge

$$h_0 = \frac{32}{31} \exp\bigl(i\omega(64 + 4\log 32)\bigr).$$

The independent verifier reconstructs the corrected rows from the original checkpoints. The rank minor $h_0 R_{\text{base},0}^2 S_{Z,0}$ excludes zero throughout the child, and the analytic first-jet endpoint shear is now

$$K_+ = 0$$

in the complete common moving factor gauge. The next T_+ dependency is the correlated transport of this sole admissible checkpoint from $r = 31$ to the common $r = 4$ section. Its first bounded chunk is now certified: one shared 8×2 $\text{Taylor}_4(\omega) \otimes \text{dual}_\tau$ state advances

$$r : 31 \longrightarrow \frac{247}{8}$$

with generator 7315, order 96, tail 0.00742651, and width 1.78454. An independent direct 16-state calculation agrees coefficientwise, the frozen spin-one tangent stays exactly zero, and the typed rank-three frame is preserved. A monolithic one-unit continuation did not finish within 180 seconds. Therefore $B_+|_{r=4}$, the explicit T_+, and a common-section numerical Stokes audit remain open;

the exact transport-free Stokes theorem above is unaffected. The next rail must resume from this checkpoint in bounded high-order chunks. The first such replay-free successor advances

$$\frac{247}{8} \longrightarrow \frac{123}{4}$$

in 25.80 seconds with order 96, tail 0.00167831, and width 2.51500. Its direct 16-state boundary comparison agrees exactly in coefficients and by interval containment, and it emits the successor payload hash `42170721...e196`. This validates the bounded checkpoint architecture but is still only one radial child, not $B_+|_{r=4}$. A second adaptive successor uses a larger certified step and advances

$$\frac{123}{4} \longrightarrow \frac{979}{32}$$

in 38.20 seconds. Its order is 120, tail 2.80109×10^{-4}, and width 3.69395; the same direct boundary gate passes. Thus the bounded driver is beginning to increase radial progress. A proposed 7/32, order-168 successor then reaches the 42-second execution cap without output. It emits no checkpoint and proves no branch or tail bound; the last usable state remains the chunk-02 payload `047f7f...b7ff`. The common section at $r = 4$ therefore remains distant. Complex-frequency work has excluded the entire growth half-plane, but no damped QNM root, contour moment, or local Smith branch has been computed. At the QNM seed disk, the direct inverse-r forward recurrence also fails as an all-order infinity remainder: at $R = 45$ its scaled direct gain is already > 1 from order 49 onward. Exterior complex scaling at angle $\pi/4$ supplies the replacement. The analytic inverse-tortoise branch from $r = 45$ is now certified to remain in $\Re r \geq 45, \Im r \geq 0$, and the scalar spin-one and spin-two Volterra norms are bounded respectively by 0.306682 and 0.310263. The resulting coarse reduced outgoing Jost balls exclude zero at the seed radius. Inward complex transport exists analytically, but a one-piece Grönwall estimate from $r = 45$ to $r = 4$ grows to 3^{69} and produces unusable 10^{33}-scale matching balls. The next quantitative rail must use panelwise centered/Lohner transport and construct the correlated τ-derivative initializer. The canonical projective tangent initializer is now bounded by $|v_\tau(45)| \leq 0.041595$ and $|\partial_x v_\tau(45)| \leq 0.001458$, but its conversion to the factor-frame coefficient b still needs the analytic endpoint gauge. A 16-panel order-16 complex-ball transport improves the global bound but still widens past 10^{17}. Affine Lohner/Taylor-model or Riccati/Grassmannian transport is therefore required. The corresponding typed projective rail has now been emitted explicitly. It certifies the generic Riccati laws for v, v_τ, v_ω, the outgoing reduced-phase chart $q = (\partial_x P_{\text{out}})/P_{\text{out}}$, and the refined panel-zero pivot lower bound 0.958711. One correlated order-26 step

$$r : 45 \longrightarrow 899/20$$

passes with remainder radii 9.61×10^{-5}, 1.88×10^{-5}, and 2.25×10^{-3} for q, q_τ, q_ω, respectively. This removes the homogeneous amplitude from the propagated state. A typed successor constructs the moving-phase horizon chart $q_H = (\partial_x P_H)/P_H$, certifies $\dot{\lambda}_H = 0$, a pivot lower bound 1.0000000974, and one finite correlated horizon step. At $r = 32$, reassembly from the content-addressed common-affine exports with the identical panel-zero generator gives

$$\Delta = q_H - q_+ + 2i\omega, \qquad \Delta_\tau = q_{H,\tau} - q_{+,\tau}, \qquad \Delta_\omega = q_{H,\omega} - q_{+,\omega} + 2i.$$

The physical mismatch again has modulus lower bound $1.7797152291 \times 10^{-4}$. Its tangent enclosures are finite but do not exclude zero, so no sensitivity or Smith conclusion is drawn. The mixed four-state tangent frame, the other 511 contour panels, a closed Evans boundary enclosure, and a root count remain open. The first two-chart Riccati rail correctly removes amplitudes, but its coarse norm-only seed loses chart separation near $r \simeq 43.3$: the v_x/v enclosure contains zero while the reciprocal chart cannot yet be certified. This is an enclosure obstruction, not a physical Jost zero. A centered order-16 reduced-amplitude initializer now constructs (v, v_τ, v_ω) on all contour panels, proves the amplitude denominator nonzero, and certifies the first inward projective segment. A rectangular Riccati continuation reaches $r \simeq 40$ before its quadratic majorant loses positivity. A midpoint-recentered affine successor now retains the shared parameter dependence of $(q, \eta, \xi) = (v_x/v, \partial_\tau q, \partial_\omega q)$ and certifies all 16 contour panels from $r = 45$ to the common radius $r = 32$. At that radius the remainder radii lie in the ranges

$$0.02603 \le r_q \le 0.18090, \qquad 0.006735 \le r_\eta \le 0.07841, \qquad 1.466 \le r_\xi \le 16.873.$$

The broad but valid ω-sensitivity is not yet sufficient for a simple-root gate. Continued inward, the scalar Riccati self-map first loses its positive discriminant between $r = 20.5$ and $r = 29.05$, depending on the contour panel. Thus the outgoing projective line reaches a useful common match radius, while the complementary horizon line, the mismatch δ, and its root/Smith gates remain open. On the horizon side, the moving-phase QNM-band Frobenius construction independently supplies certified q, η, ξ seeds on all 16 panels and again gives $\dot{\lambda}_H = 0$. Its first affine outward continuation refuses at the reference-trajectory Riccati discriminant, at $r \simeq 2.672868$ on panels 0–7 and $r \simeq 2.714923$ on panels 8–15. At that obstruction the full-panel q enclosures nevertheless exclude zero, with certified modulus lower bounds between 0.21078 and 0.24612. The reciprocal chart

$$p = q^{-1}, \qquad p_\tau = -q_\tau/q^2, \qquad p_\omega = -q_\omega/q^2$$

is therefore admissible on every panel. A logarithmic-norm successor carries all 16 horizon lines to $r = 4$, accepting 129 or 133 steps with no rejected trial. A checkpointed continuation then reaches $r = 8, 16, 32$ on all panels, with 680 accepted steps per panel, no rejected trial, and no second chart change. At $r = 32$ the reciprocal-base radii lie between 0.104 and 0.205, the τ-sensitivity radii between 1.06 and 7.65, and the ω-sensitivity radii between 1.26 and 4.62. Thus the two selected endpoint lines now reach the same projective matching radius. Because their reduced amplitudes carry opposite oscillatory phases, the physical common-chart mismatch is

$$\Delta = q_H - q_+ + 2i\omega, \qquad \Delta_\tau = q_{H,\tau} - q_{+,\tau}, \qquad \Delta_\omega = q_{H,\omega} - q_{+,\omega} + 2i.$$

This corrected mismatch has now been assembled on all 16 panels. Its midpoint magnitudes, 4×10^{-5} to 2×10^{-3}, exhibit the expected near-resonant cancellation. The separately serialized endpoint remainders do not retain a common cross-endpoint affine generator, however, so their rigorous radii must be added and lie between 0.304 and 0.393. Every boundary panel then contains zero. The boundary-nonvanishing gate therefore refuses with `SERIALIZED_CROSS_ENDPOINT_DEPENDENCY_WIDTH_CONTAINS_ZER` the argument-principle and defect-moment gates are deliberately not run. An export audit makes the required repair contract explicit. Both endpoint artifacts must carry the same frequency-generator identity, the centered polynomial coefficients of q, q_τ, q_ω, residuals enclosed only after subtraction of those polynomials, and an explicit reduced-phase convention tied to that generator. The present serialized balls do not supply those fields. A bounded singleton-frequency prototype passes its first horizon Taylor reference but refuses immediately at the horizon q-remainder self-map; no long joint rerun or downstream contour gate is therefore inferred. A first implementation of the complete common-affine export contract has now made the phase rule $\Delta = q_H - q_+ + 2i\omega$ and one panel-local generator identity explicit. Adaptive halving repairs the outgoing singleton self-map, while scaling the reciprocal horizon step by $(r-2)/16$ repairs its near-horizon discriminant. Both box and singleton endpoint polynomial exports then reach $r = 32$ on panel zero of the 512-panel contour under the common generator `qnm-boundary-omega-0000-of-0512`. The physical boundary gate still initially refuses because its independent residual radius is 0.0335412. Replacing the singleton endpoint transports by order-26 direct-q transport with adaptive radial recentering reduces the combined residual to 3.25322×10^{-5}. The physical mismatch then has certified modulus lower bound

$$1.77972 \times 10^{-4} > 0$$

on panel zero. This is the first strict common-affine Evans-boundary gate, but only for one of 512 panels. A checkpointed four-worker successor now extends the same panel-local shared-generator construction through panel 76. All 77 physical mismatch enclosures on this contiguous prefix exclude zero; the smallest modulus lower bound is 9.49348×10^{-6}, on panel 76. This remains a strict boundary segment, not a closed contour. Typed Riccati audits independently emit $(\Delta, \Delta_\tau, \Delta_\omega)$ on every co-located panel. Every current rectangular Δ_τ and Δ_ω ball contains zero, so neither the intrinsic selector nor the simple-root Newton denominator is certified. The initial panel-77 refusal was traced to cancellation in the subtractive binary64 formula for the smaller quadratic self-map root, not to a loss of the Riccati enclosure. The stable formula

$$\frac{2q_c}{-q_b + \sqrt{q_b^2 - 4q_a q_c}}$$

with an exact 1000001/1000000 outward enlargement proves the unchanged strict self-map margin 8.62645×10^{-31}. Both reciprocal pivots exclude zero, and panel 77 then has

$$|\Delta| \geq 8.86836 \times 10^{-6}.$$

The repaired rail then certifies twenty further panels without changing any threshold. The certified contiguous boundary prefix is therefore panels 0–97; its smallest modulus lower bound is

$$2.47379 \times 10^{-7}$$

at panel 97. The parent enclosure for panel 98/512 contains zero even though both endpoint transports pass. Exact dyadic subdivision repairs this without changing any threshold: the children 196/1024 and 197/1024 have modulus lower bounds 2.55042×10^{-5} and 2.53795×10^{-5}, respectively. Two subsequent bounded adaptive chunks apply the same rule to parents 99/512 and 100/512. Their four dyadic children all exclude zero. A third bounded continuation repairs parents 101/512 and 102/512 with the four children $202/1024, \ldots, 205/1024$, extending the exact contiguous boundary coverage through 103/512. Parent 103/512 is the next gap. Its coarse enclosure has now been evaluated and content-addressed, but it fails; the bounded invocation ended before the two required children 206/1024 and 207/1024 could be launched. A child-only successor then certifies those two intervals with modulus lower bounds 2.44667×10^{-5} and 2.43794×10^{-5}. Exact coverage therefore reaches $104/512 = 13/64$. Parent 104/512 likewise fails closed at coarse resolution; its dyadic children 208/1024 and 209/1024 exclude zero with modulus lower bounds 2.42977×10^{-5} and 2.42106×10^{-5}. Exact contiguous coverage now reaches 105/512, with 105/512 the next gap. The unchanged rail then repairs parents 105/512 and 106/512 through children $210, \ldots, 213$ at denominator 1024. A fixed 1/1024 continuation evaluates and accepts the next 32 ordered panels $214, \ldots, 245$, with mismatch lower bounds remaining above 2.32×10^{-5}. Exact contiguous coverage now reaches

$$\frac{246}{1024} = \frac{123}{512},$$

with 246/1024 the next gap. The 32-panel producer took 108.81 seconds, so it is retained as an affected-chain exhaustive run with a fast replayless verifier; subsequent compute chunks must be smaller. The first bounded successor chunk evaluates and accepts panels $246, \ldots, 253$ at denominator 1024 in 27.94 seconds. Exact contiguous coverage therefore reaches

$$\frac{254}{1024} = \frac{127}{512},$$

with 254/1024 the next gap. The remaining boundary, interval-Newton, argument-principle, and Smith/EP2 gates are not run. Uniform time-domain boundedness, decay, and completeness remain open.

The present result already fixes an important baseline:

> Additional curvature directions occur in the future-horizon carrier, and independently the formal infinity modules contain finite non-Einstein or mixed directions. Neither leading endpoint diagnostics nor formal radial pairing finiteness selects the Einstein kernel. More strongly, the complete three-dimensional indefinite incoming trace is globally populated by future-horizon-regular solutions at every real positive frequency. The complete outgoing trace is also populated throughout the certified cell $0.49995 \leq \omega \leq 0.50005$; analyticity extends population to an open dense full-measure positive-frequency set and gives dense outgoing range on every compact positive-frequency band. Explicit outgoing amplitudes, the isolated reflection-zero set and complete-pilot pointwise outgoing rank, the damped QNM spectrum, time-domain stability, CPT positivity, and a complete scattering operator remain open; exponentially growing separated axial modes are already excluded.

A Explicit axial endpoint Grams and uniform bounds

In the ordered bases

$$\mathscr{I}^- : (\mathrm{XI0}, \mathrm{XI1}, \mathrm{EI0}), \qquad \mathscr{I}^+ : (\mathrm{XI2}, \mathrm{XI3}, \mathrm{EI2}),$$

the normalized incoming past Gram is

$$G_- = \begin{pmatrix} \frac{576}{5\omega} & \frac{96i}{5} & \frac{384\omega}{5} \\ -\frac{96i}{5} & -\frac{144\omega}{5} & -\frac{192i\omega^2}{5} \\ \frac{384\omega}{5} & \frac{192i\omega^2}{5} & 0 \end{pmatrix}, \tag{43}$$

and the normalized outgoing future Gram is

$$G_+ = \begin{pmatrix} \frac{384(-512\omega^6 + 640\omega^4 - 278\omega^2 + 39)}{5\omega} & \frac{12288i\omega^4 - 3072\omega^3 - 9984i\omega^2}{5} + 192\omega + 384i & \frac{-1536i\omega^2 + 384\omega + 768i}{5} \\ \frac{-12288i\omega^4 - 3072\omega^3 + 9984i\omega^2}{5} + 192\omega - 384i & -\frac{768\omega^3}{5} + 48\omega & \frac{96\omega}{5} \\ \frac{1536i\omega^2 + 384\omega - 768i}{5} & \frac{96\omega}{5} & 0 \end{pmatrix}. \tag{44}$$

The full current Grams are $\pi\alpha_{\mathrm{W}} G_\pm$.

The exact squared Frobenius functions have certified maxima

	$\max_{\omega\in[1/2,3/4]} \lVert G_\pm \rVert_F^2$	$\max_{\omega\in[1/2,3/4]} \lVert G_\pm^{-1} \rVert_F^2$
G_-	1429056/25	7025/65536
G_+	10389888/25	19825/65536.

(45)

For G_-, both functions are strictly decreasing. For G_+, exact Sturm counts find one interior critical point for each function, and the endpoint derivative signs are $(-,+)$, proving that the critical point is a minimum. These facts prove (36) without inferring a lower bound merely from the determinants.

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